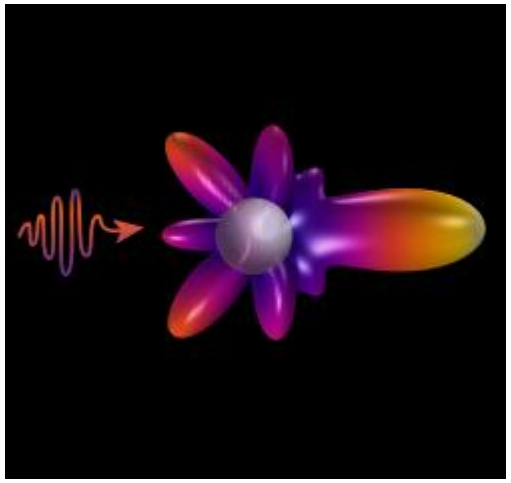
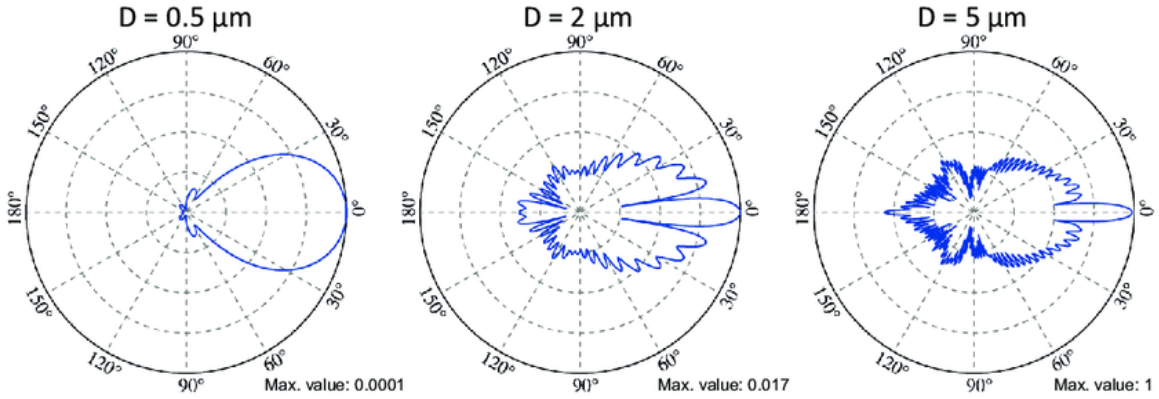
Polarimetry by single water droplet

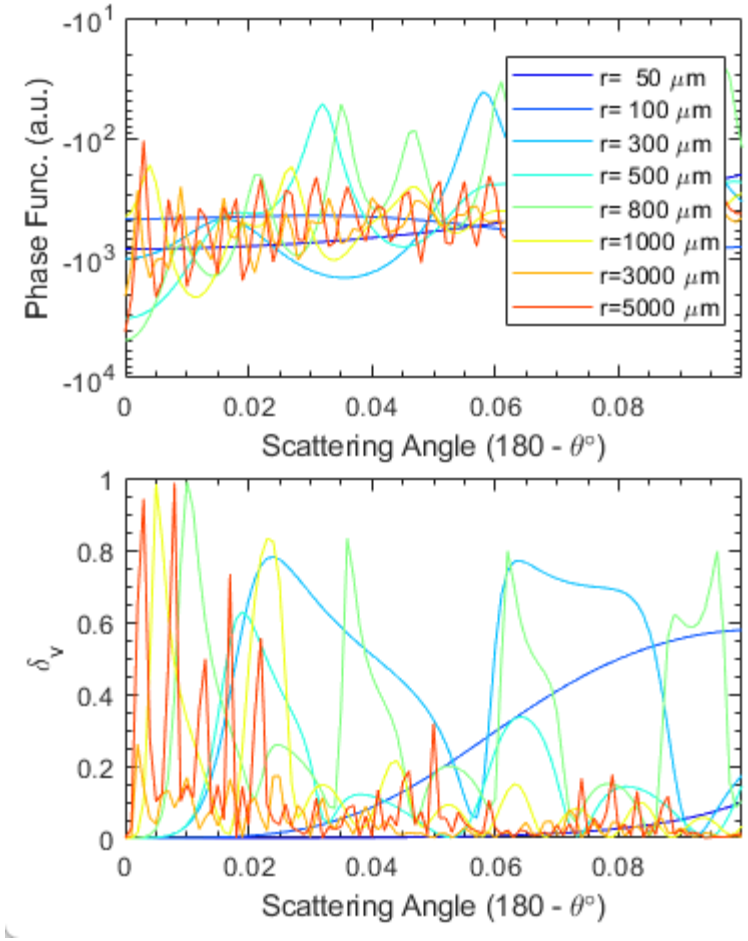
Scattering by spheric particle (Mie Scattering)





incident wavelength = 473 nm

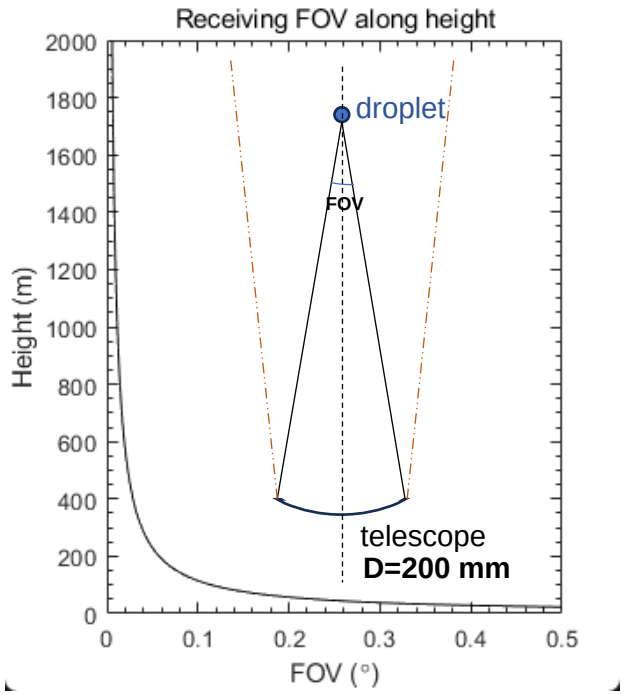
Zoom-in at near-backscatter angles



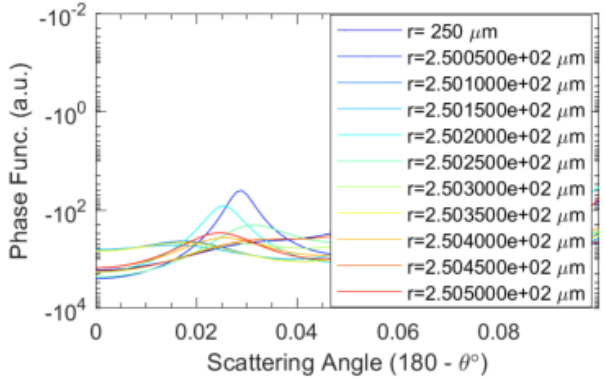
- Very complex structure at near-backscatter angles
- Strong depol. component with high sensitivity to particle size and scattering angles

Polarimetry by single water droplet

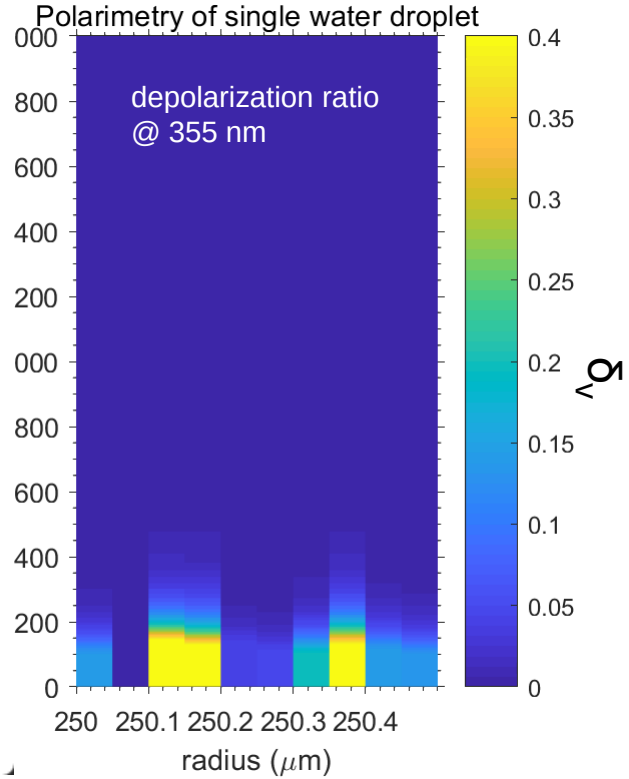
Lidar receiving geometry with height



r=250 um



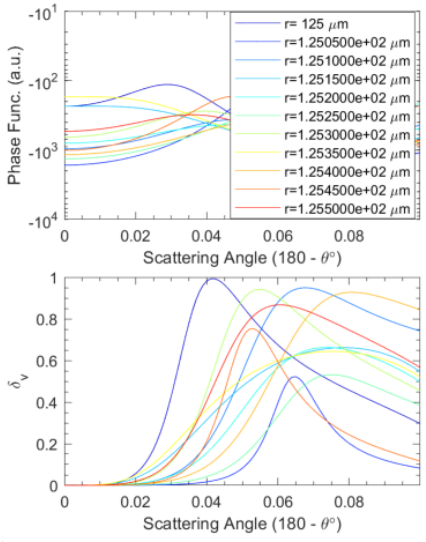
assume rain droplet with the same size

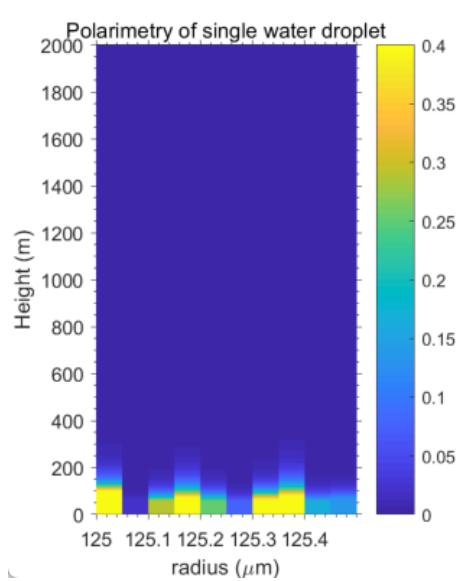


- δ_v enhancement at lower height
- peak value of δ_v fluctuates with particle size
- peak value height not sensitive to particle size

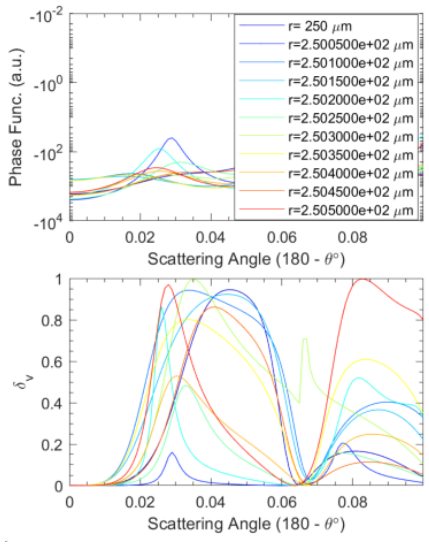
Mind the constrain of field of view of the telescope

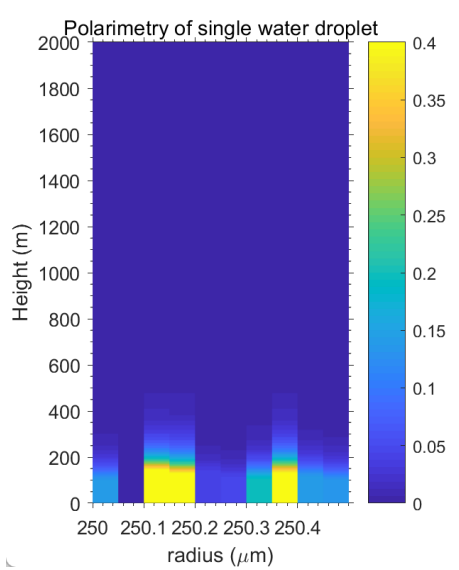
r=125 um



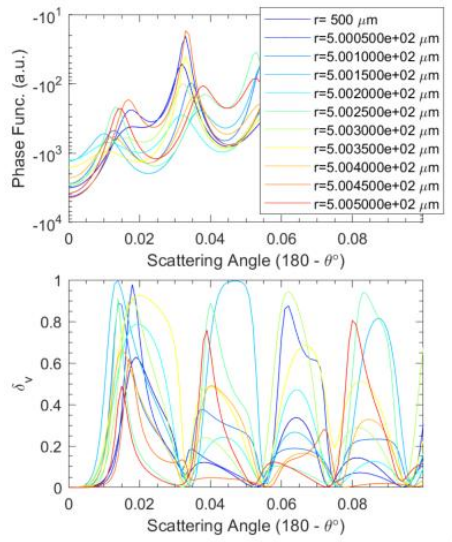


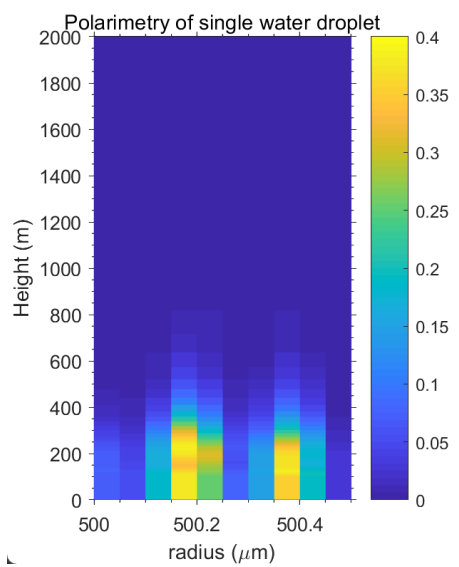
r=250 um



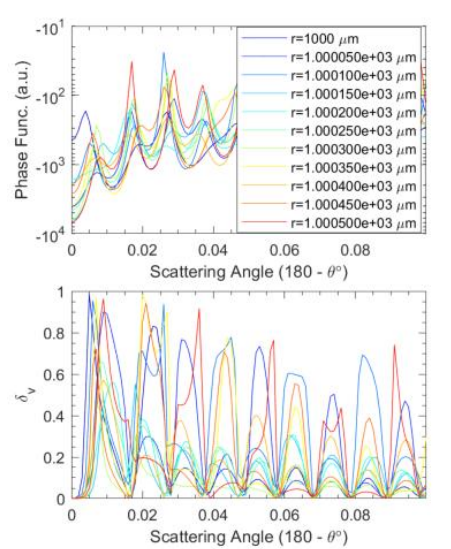


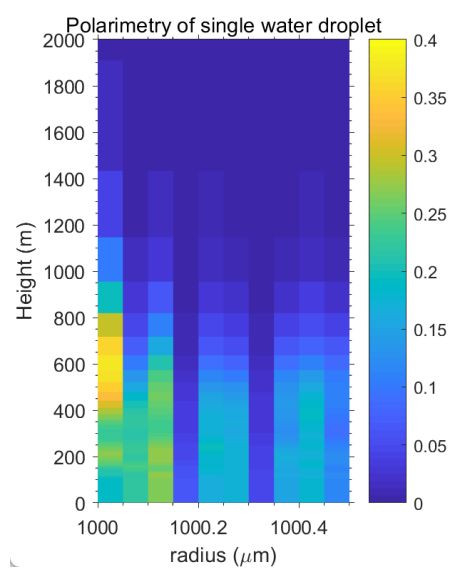
r=500 um



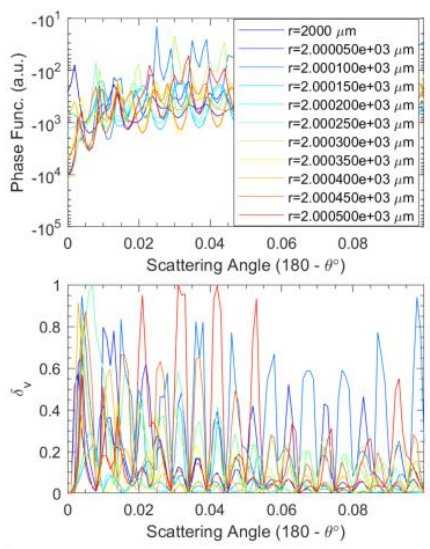


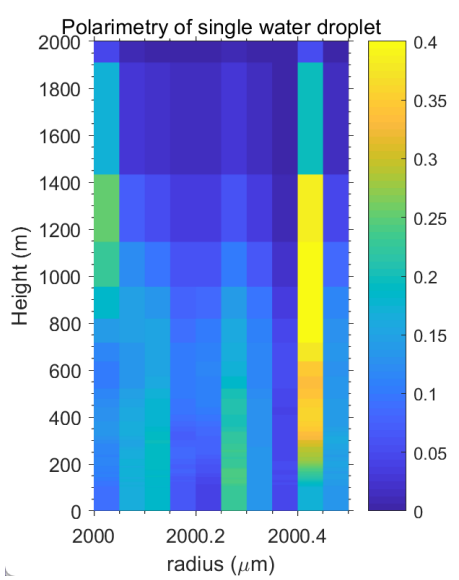
r=1000 um



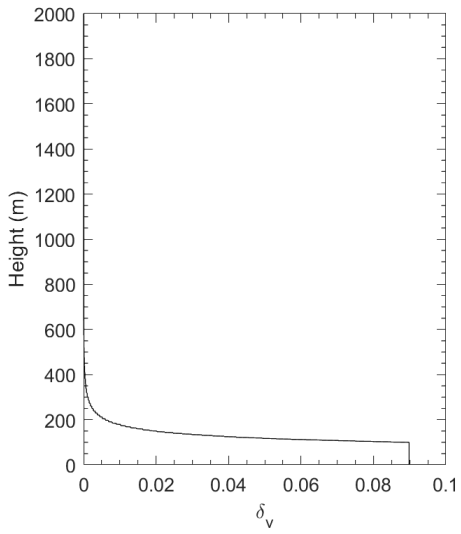


r=2000 um

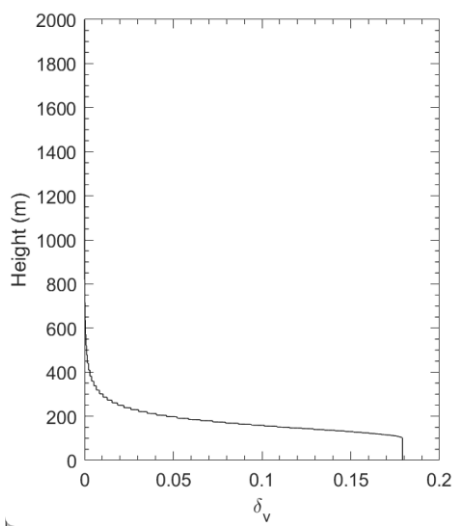




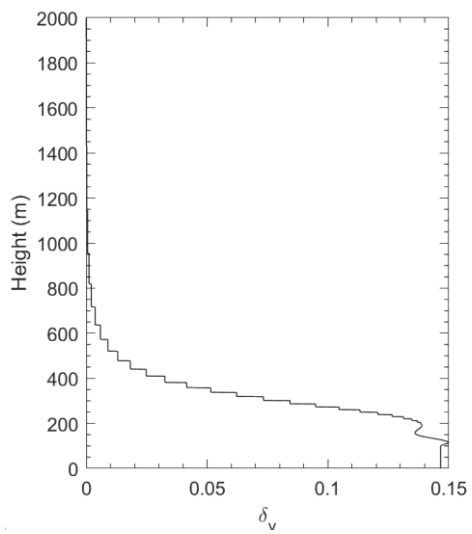
r=125 um



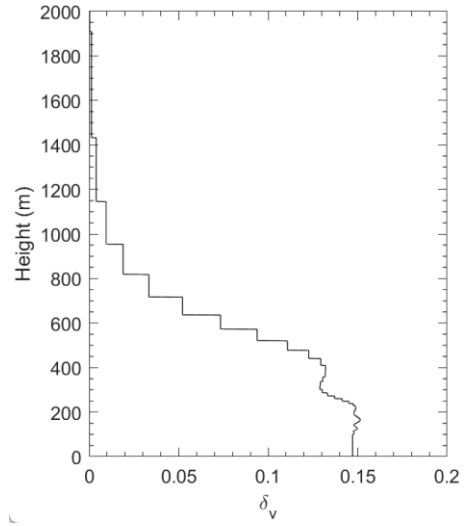
r=250 um



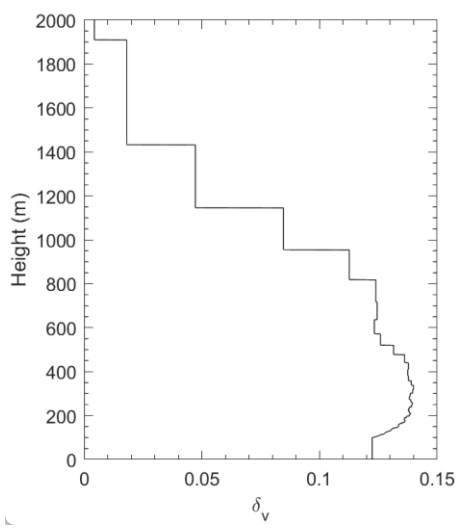
r=500 um

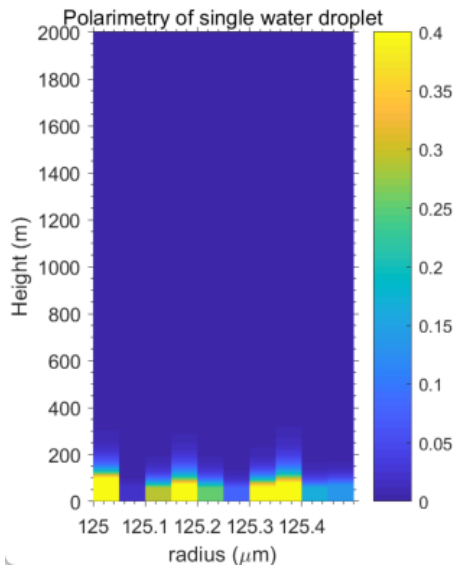


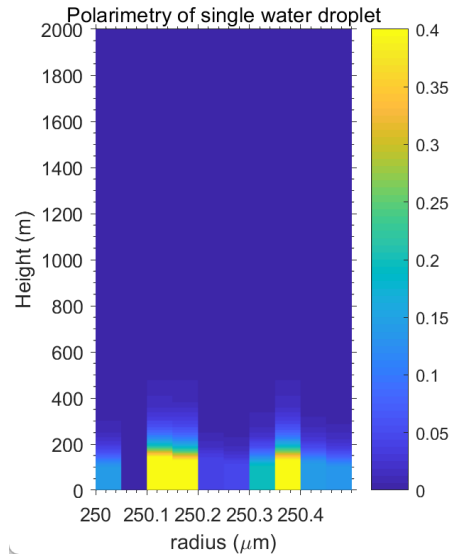
r=1000 um

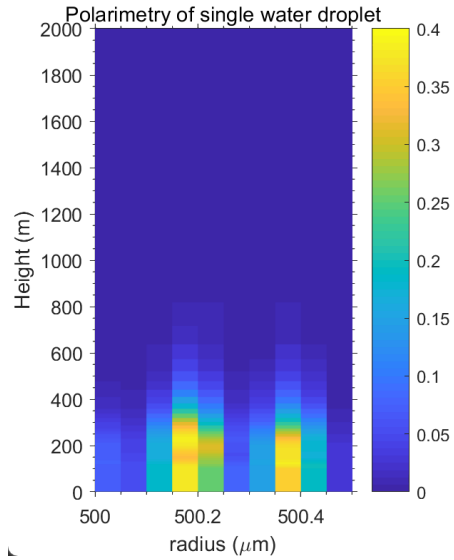


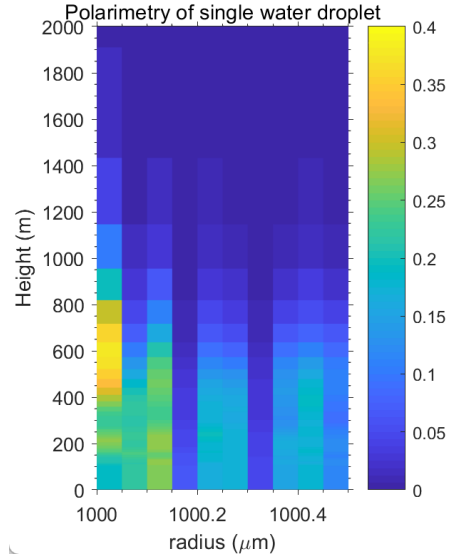
r=2000 um

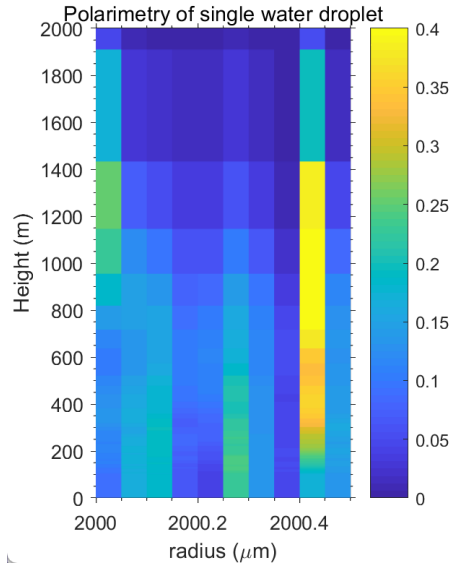






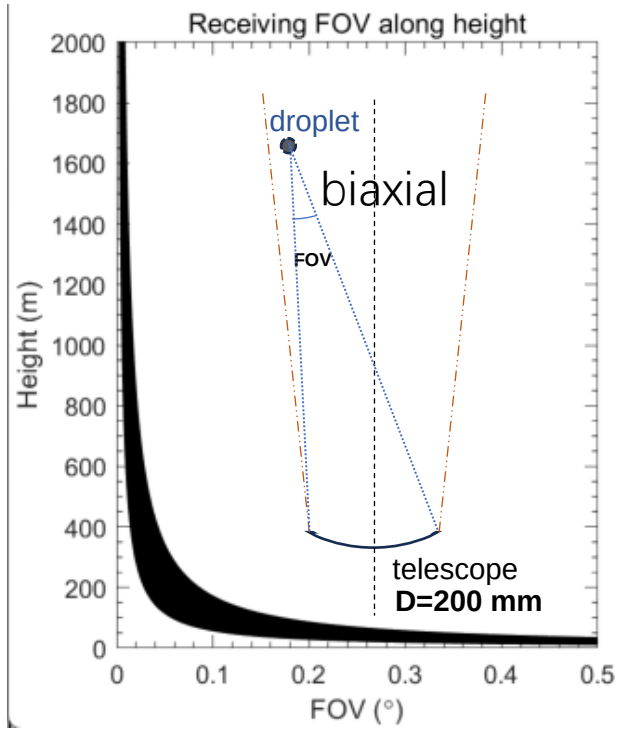






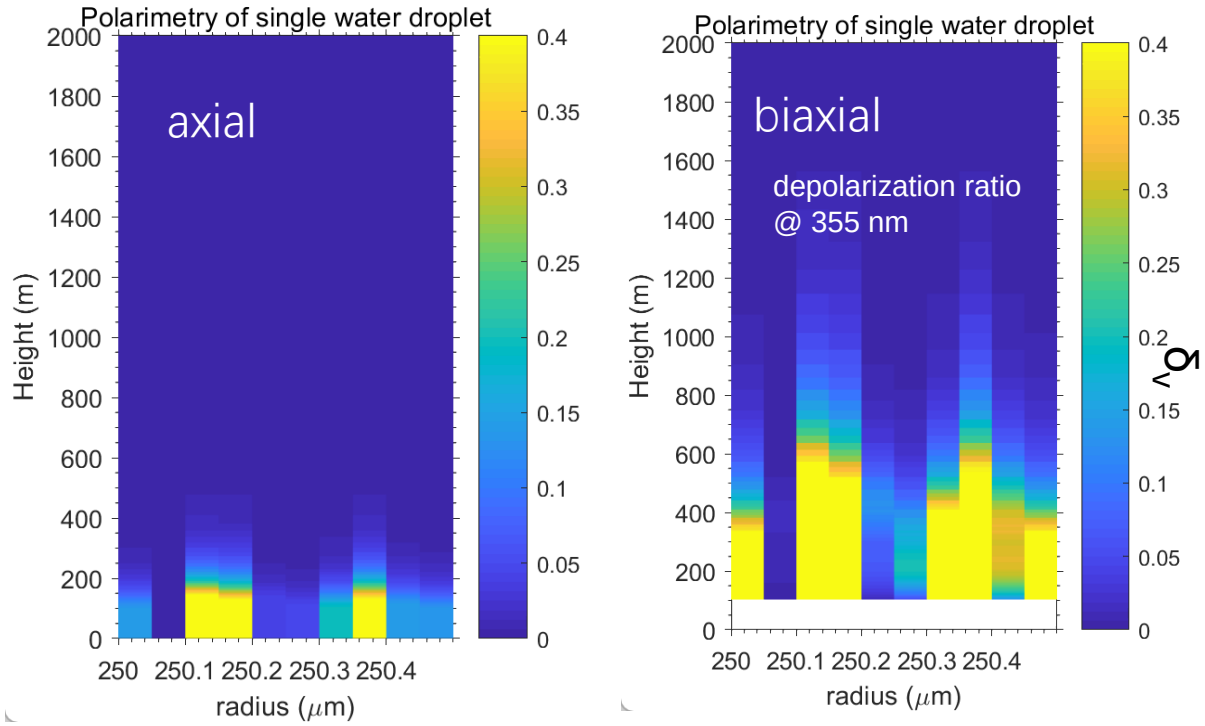
Polarimetry by single water droplet

Lidar receiving geometry with height



Assume distance = 200 mm

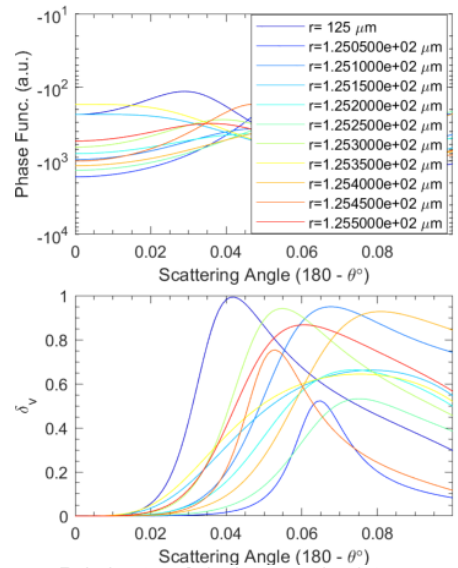
r=250 um

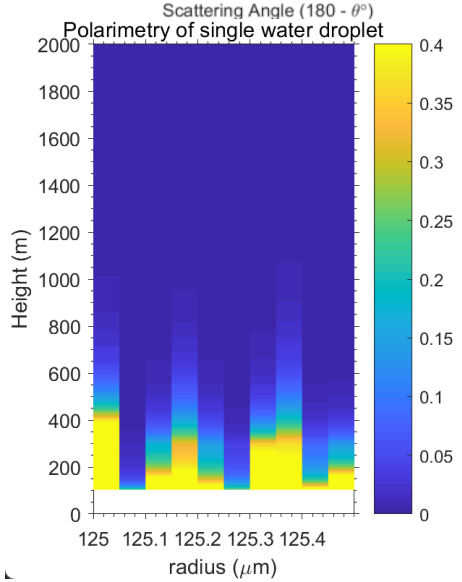


- δ_v enhancement at lower height
- peak value of δ_v fluctuates with particle size
- peak value height not sensitive to particle size

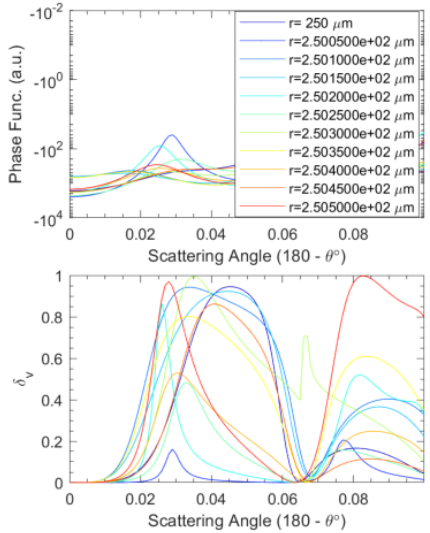
Mind the constrain of field of view of the telescope

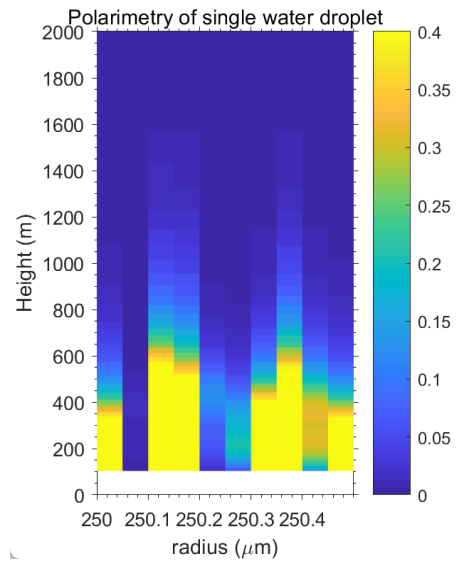
r=125 um



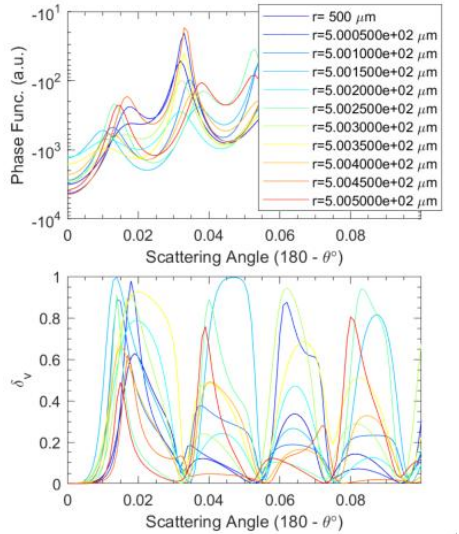


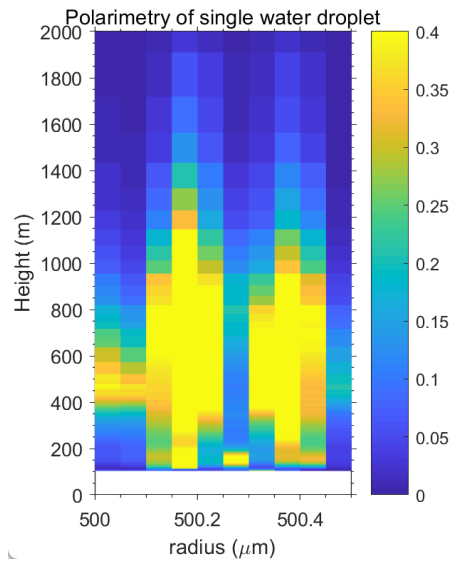
r=250 um



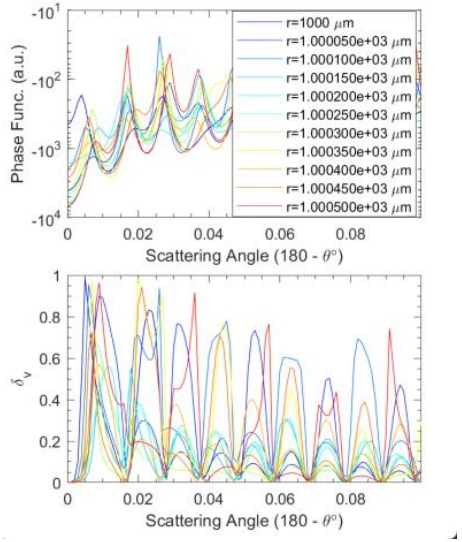


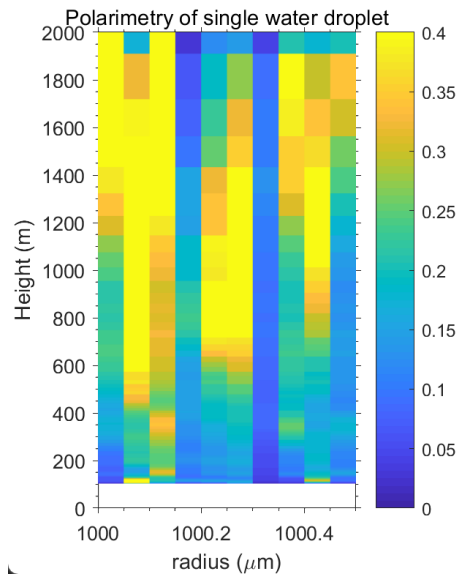
r=500 um



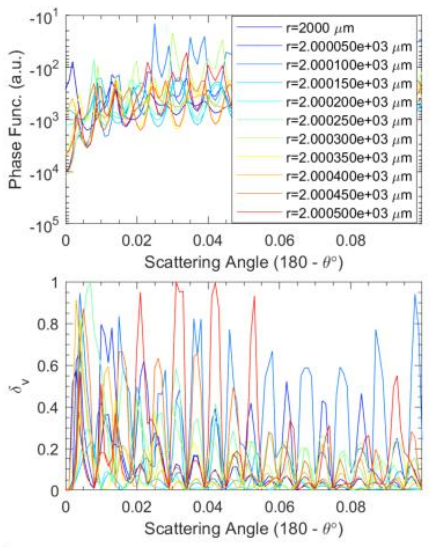


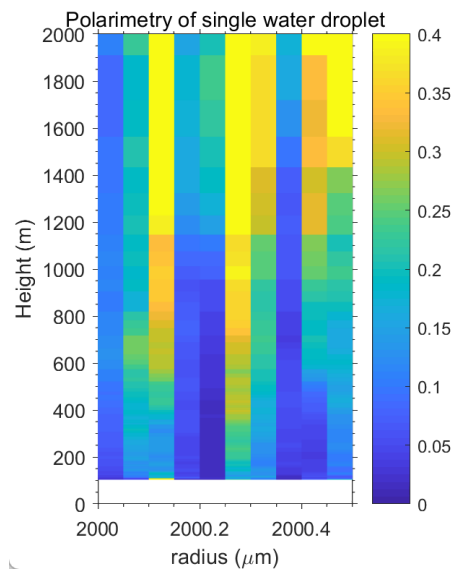
r=1000 um



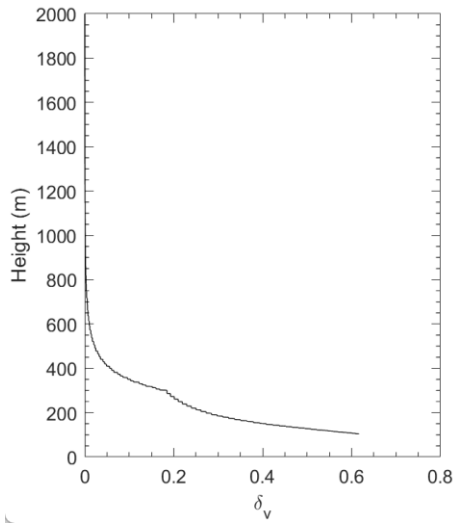


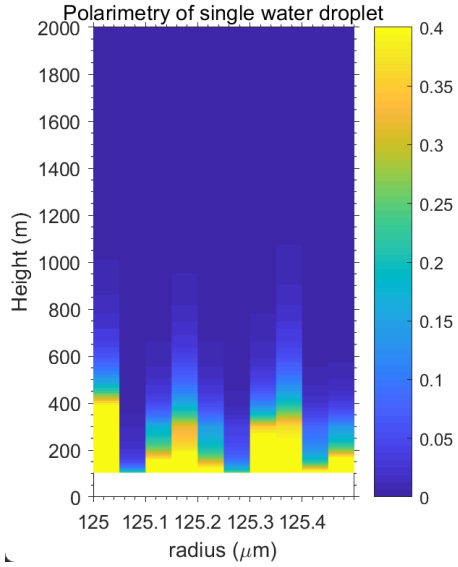
r=2000 um



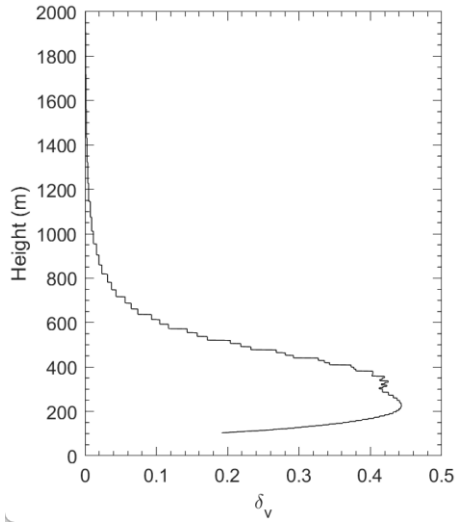


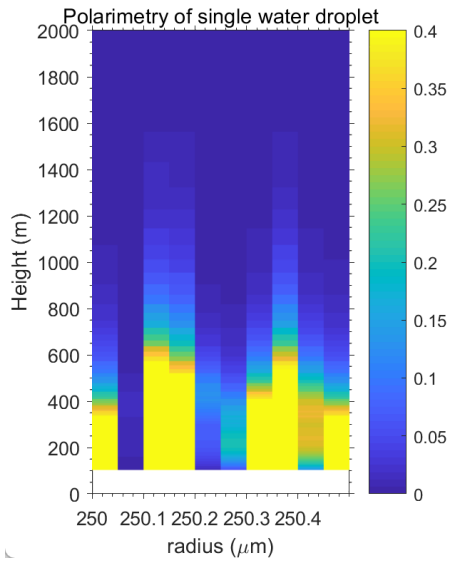
r=125 um



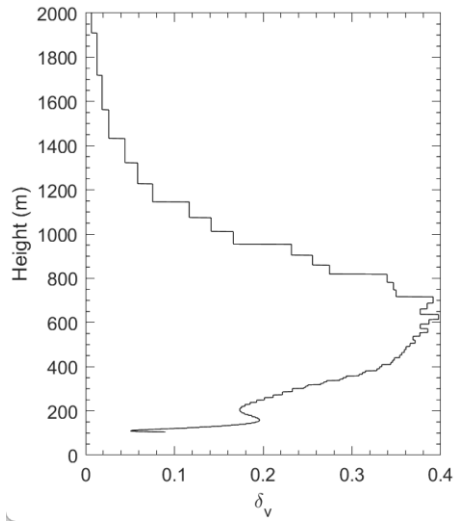


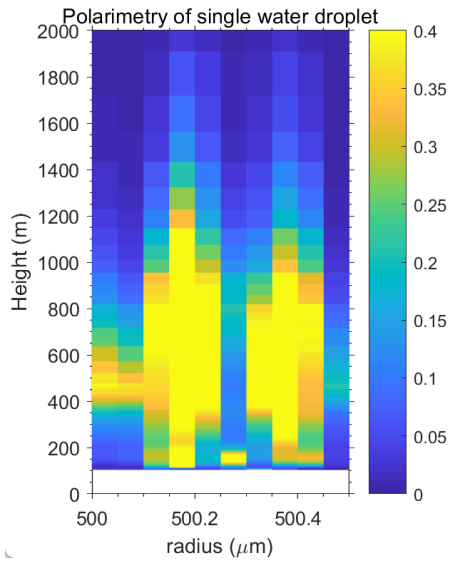
r=250 um



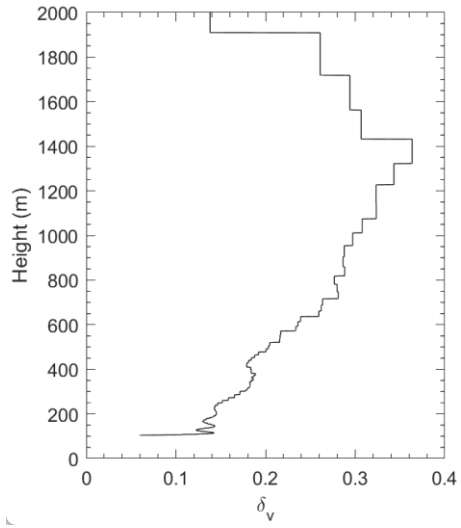


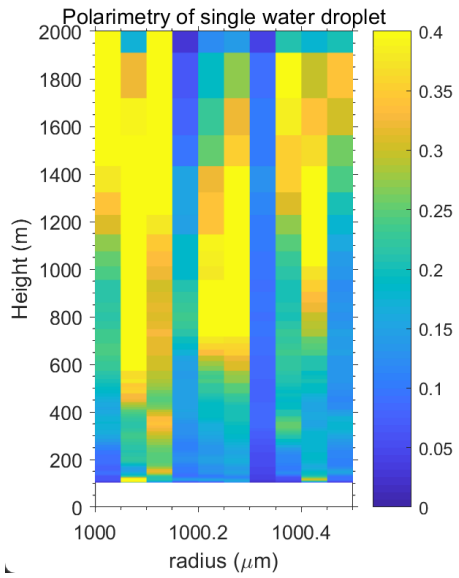
r=500 um



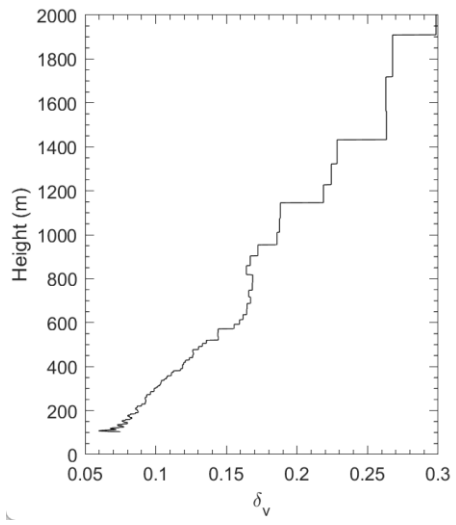


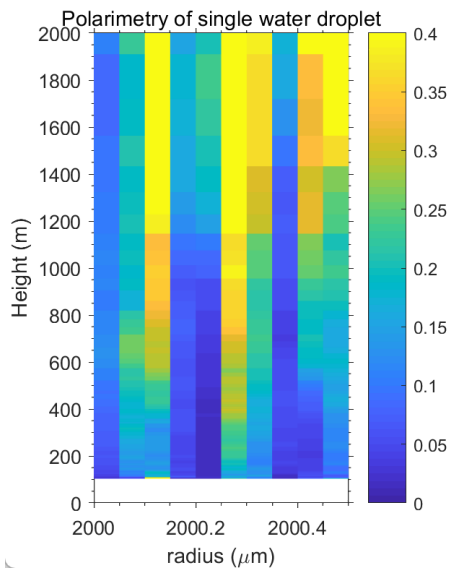
r=1000 um





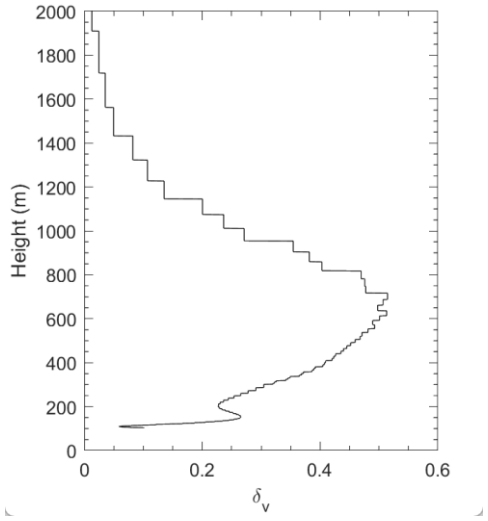
r=2000 um

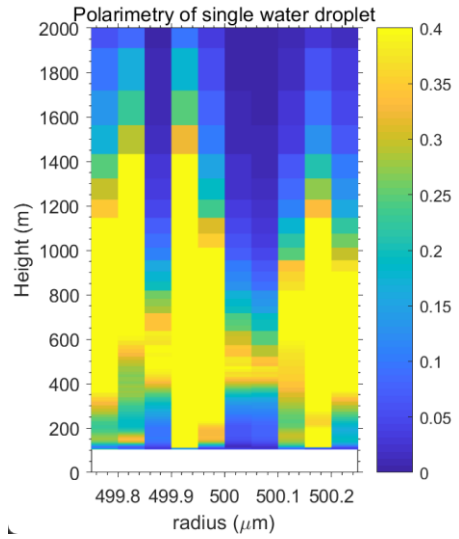




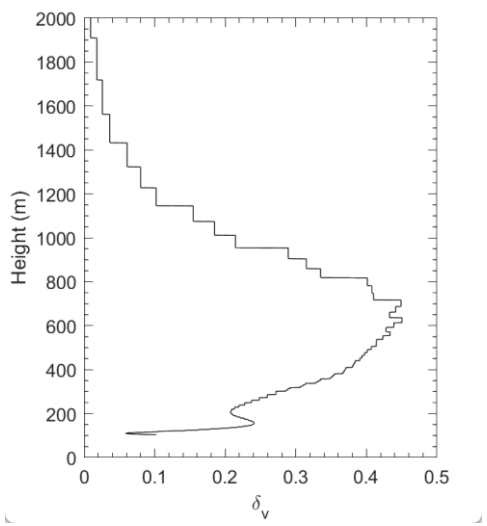
Polarimetry by water droplets with similar size distribution

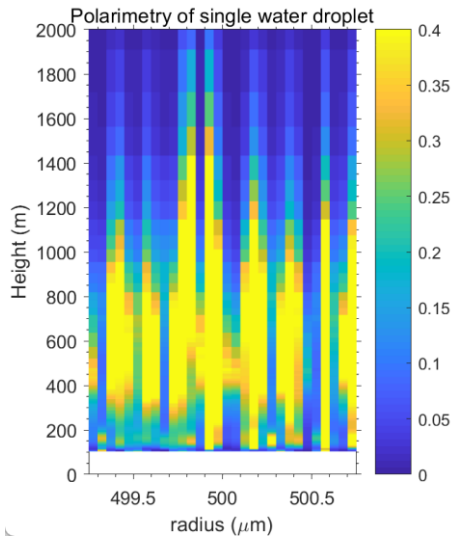
999.5-1000.5 um



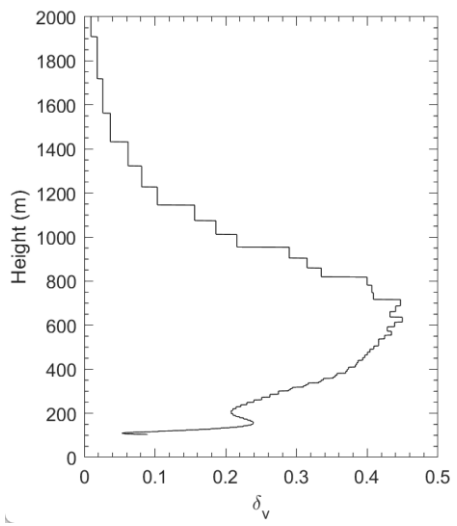


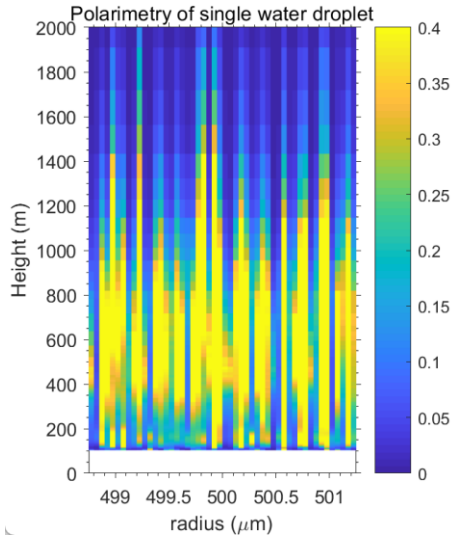
998.5-1001.5 um



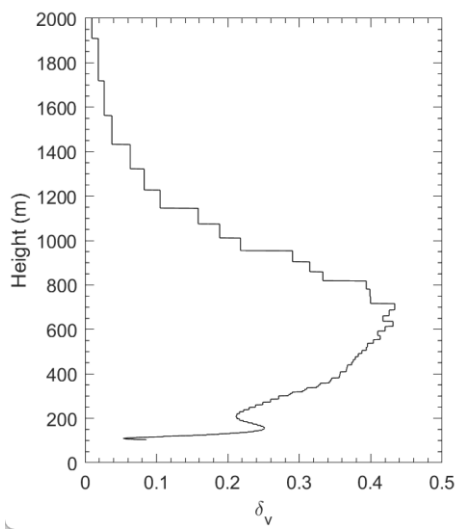


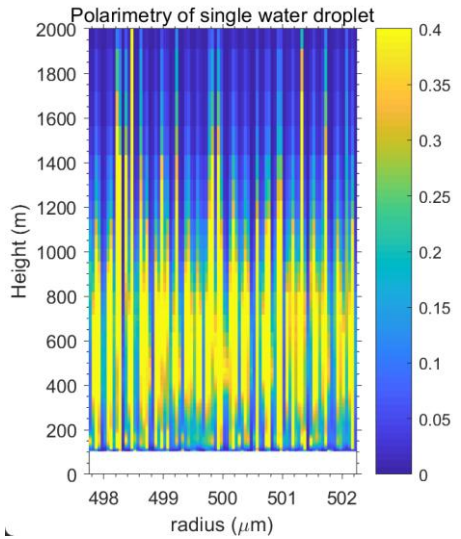
997.5-1002.5 um





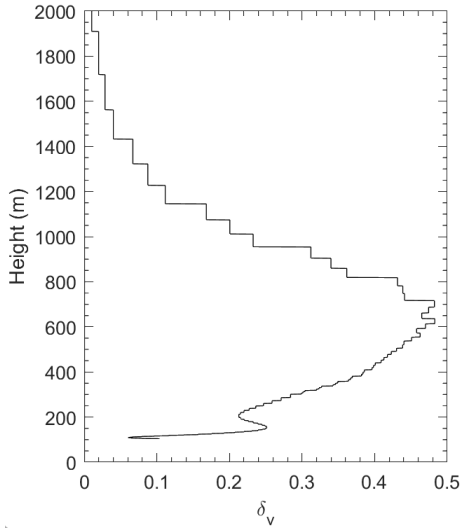
995.5-1004.5 um

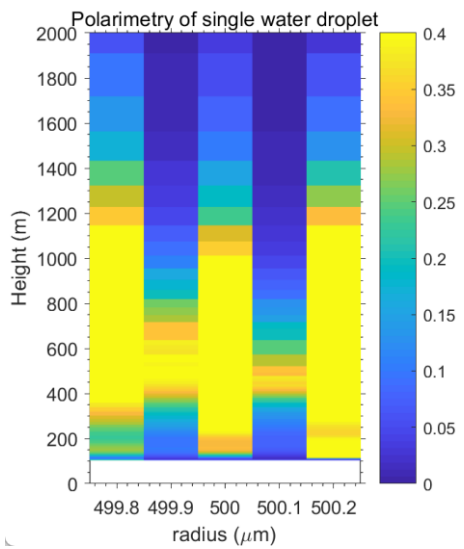




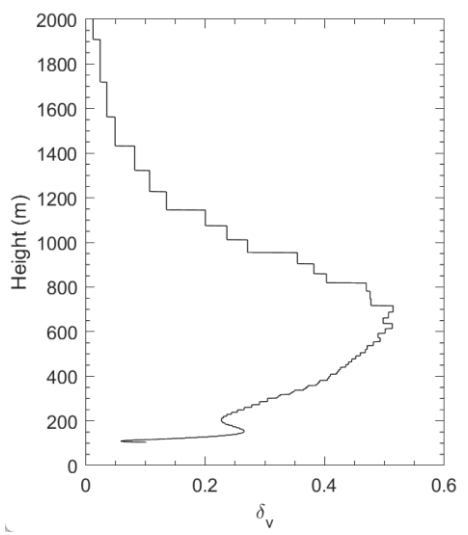
Polarimetry by water droplets with similar size distribution

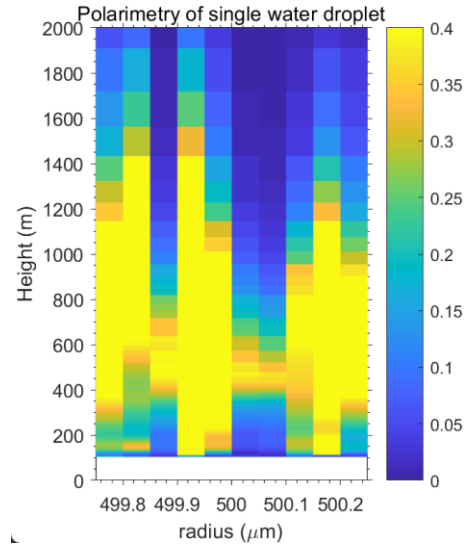
999.5:0.2:1000.5 um



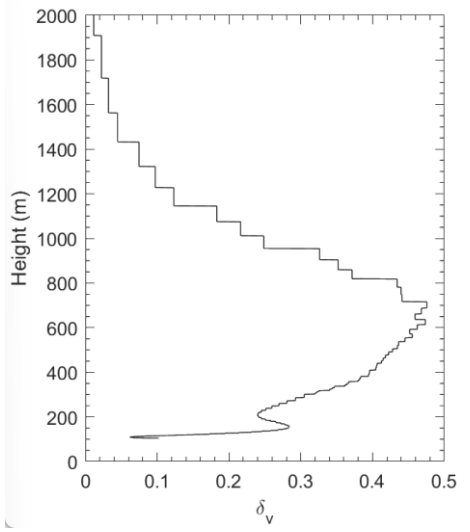


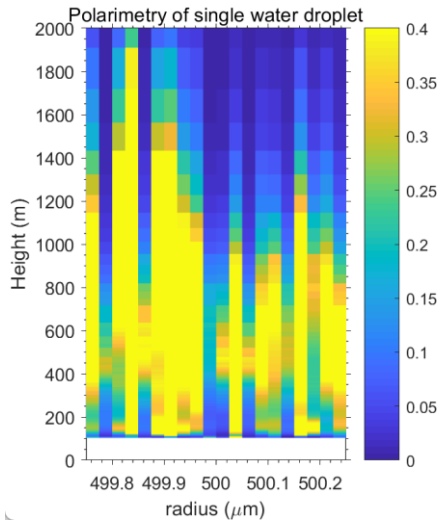
999.5:0.1:1000.5 um



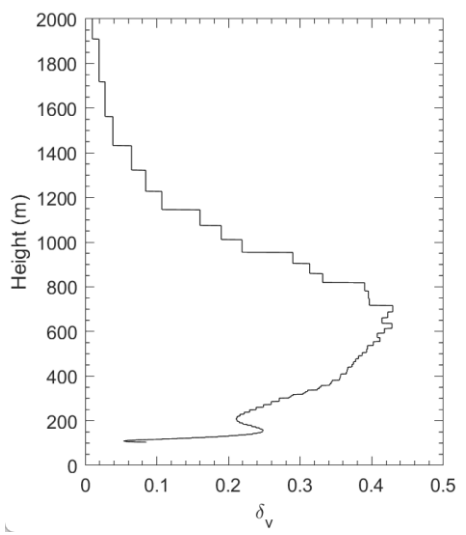


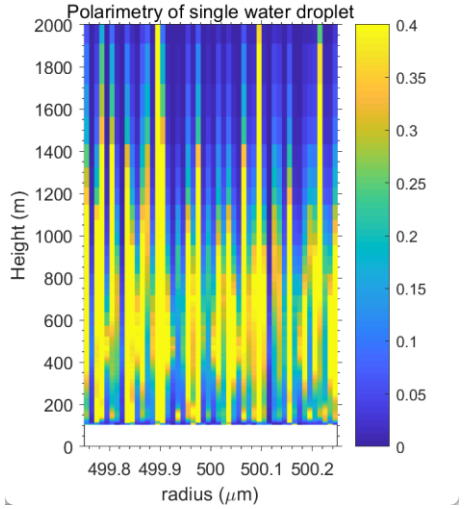
999.5:0.05:1000.5 um





999.5:0.02:1000.5 um

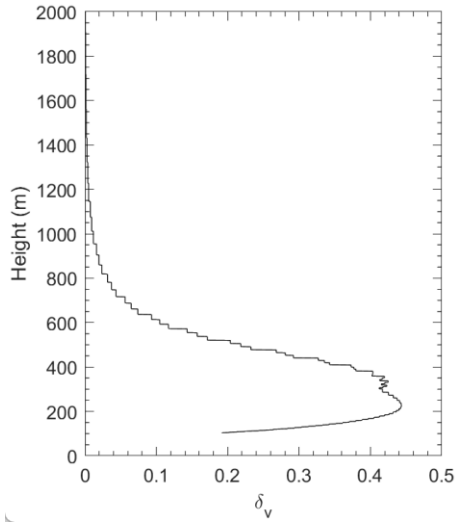


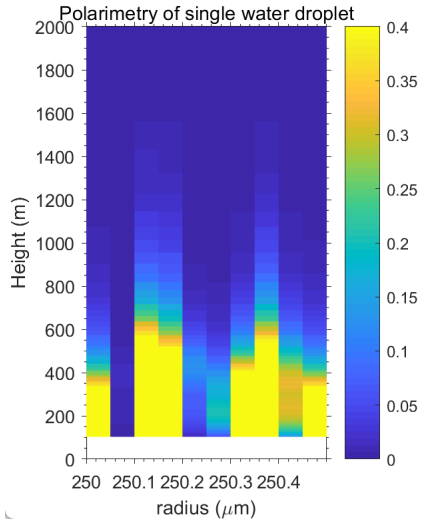


Polarimetry by water droplets with similar size distribution

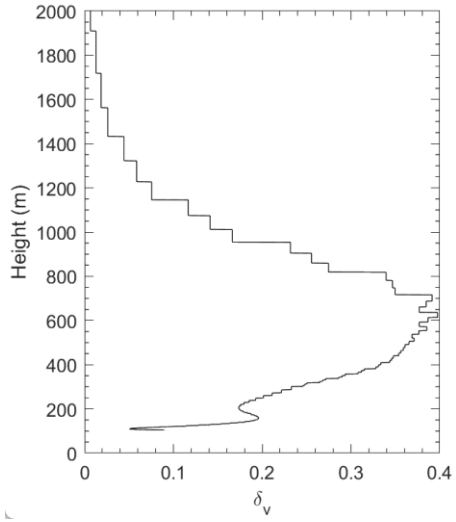
355 um

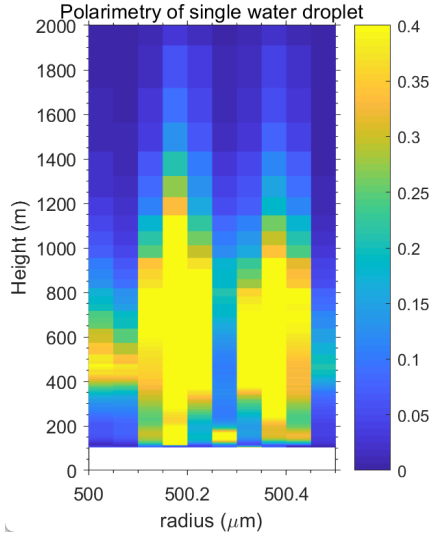
250:0.1:251 um



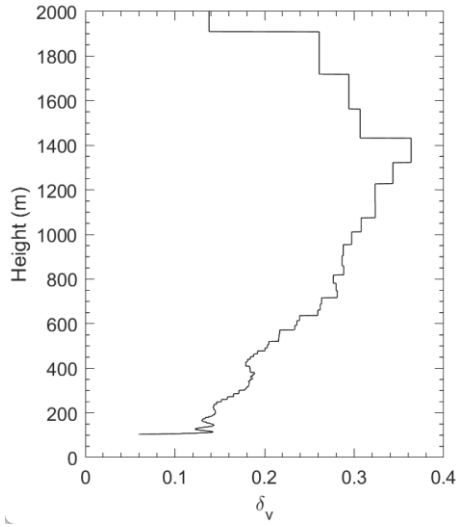


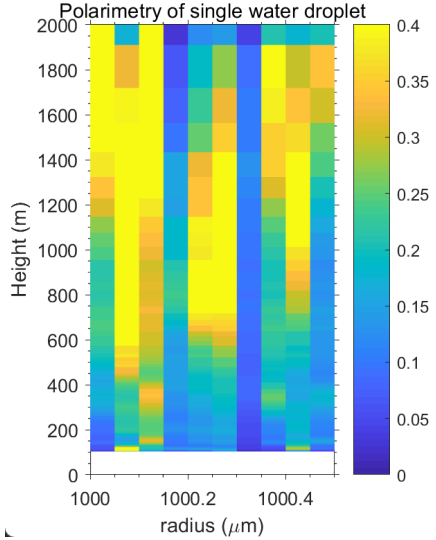
500:0.1:501 um



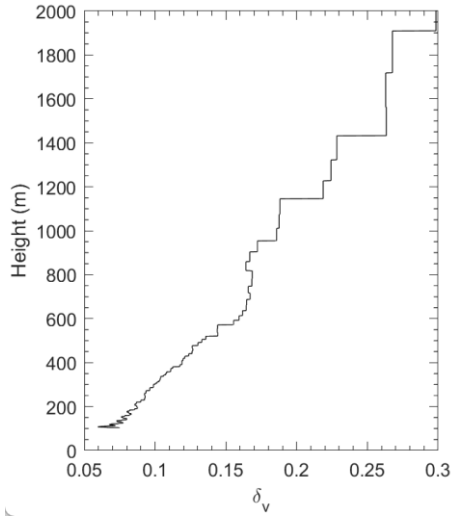


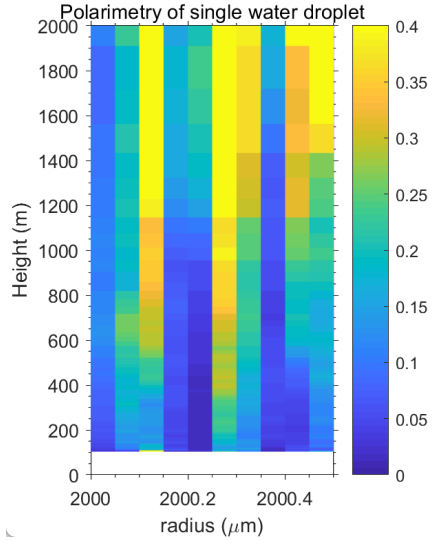
1000:0.1:1001 um





2000:0.1:2001 um

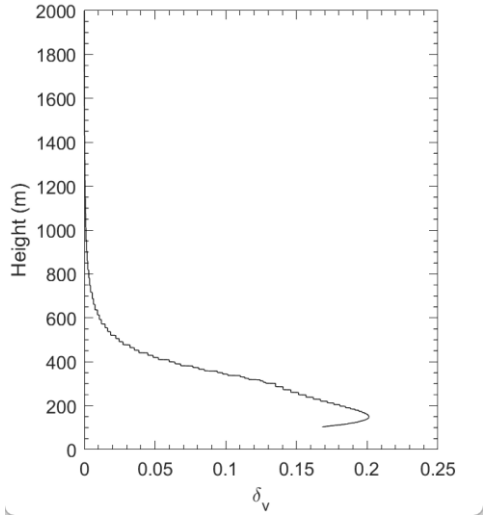


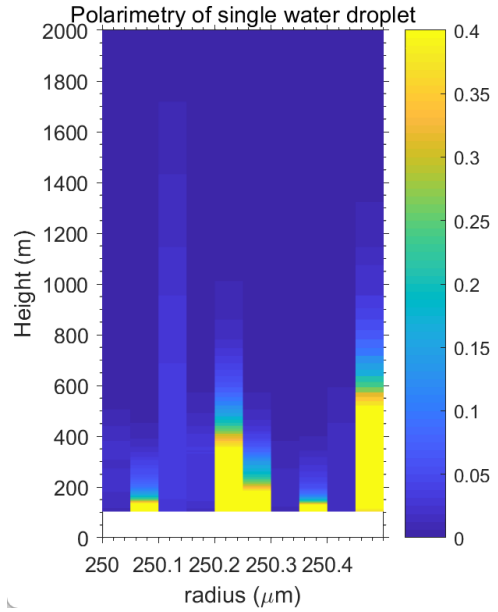


Polarimetry by water droplets with similar size distribution

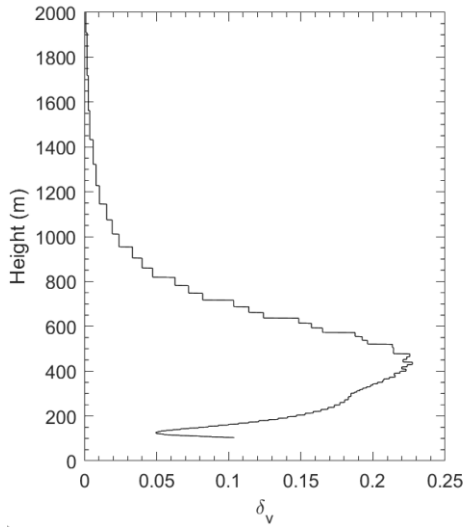
532 um

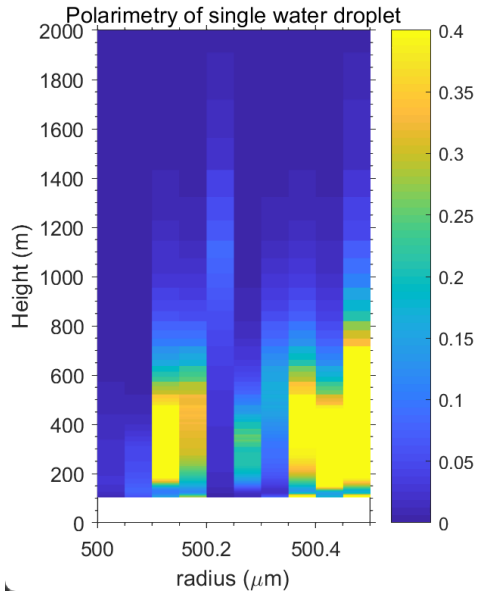
250:0.1:251 um



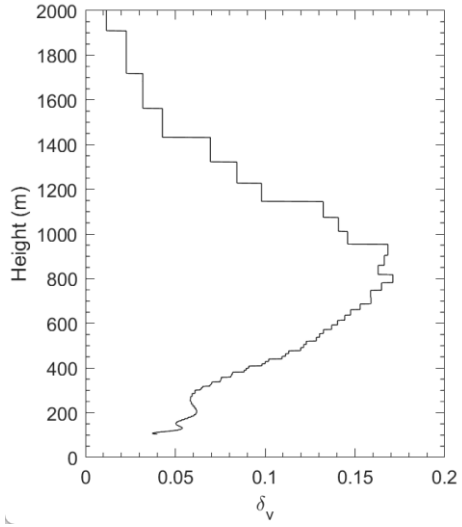


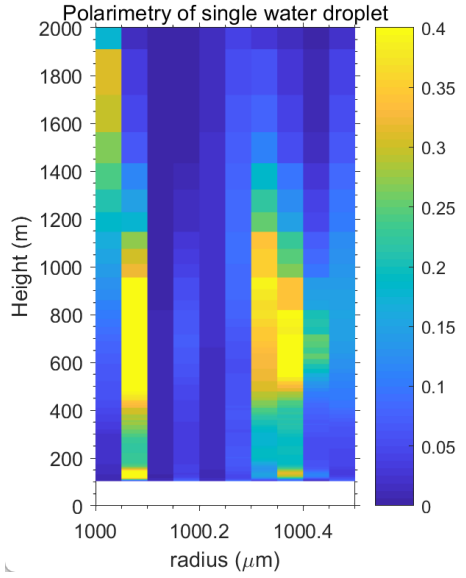
500:0.1:501 um



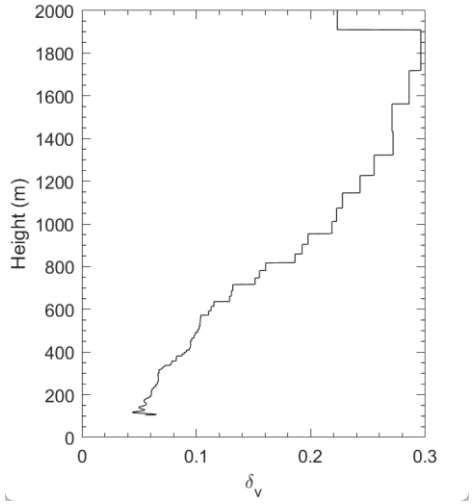


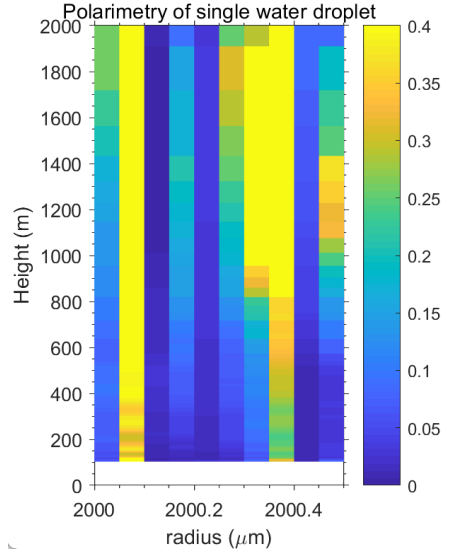
1000:0.1:1001 um



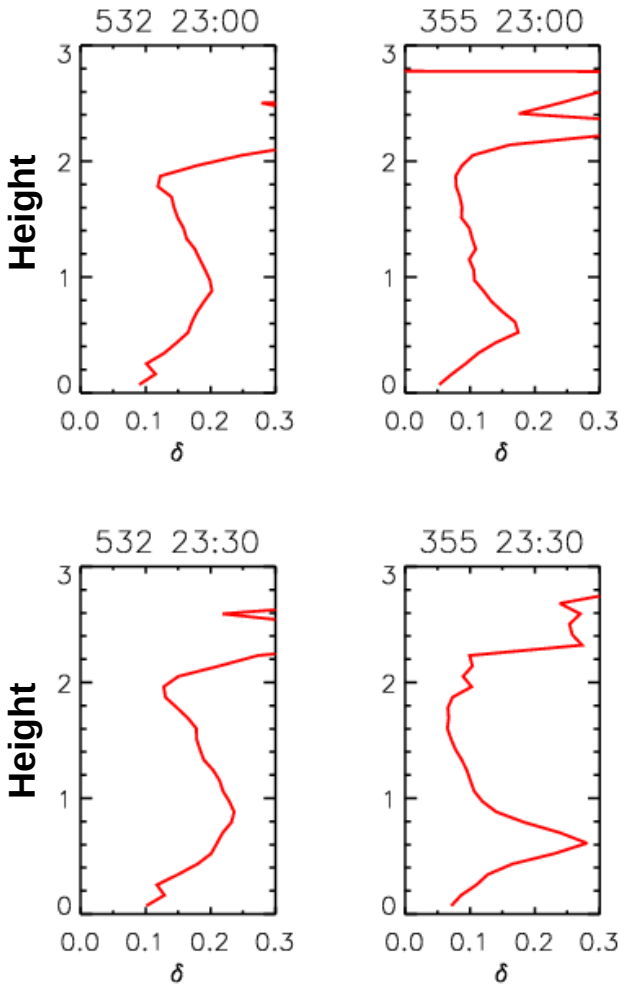


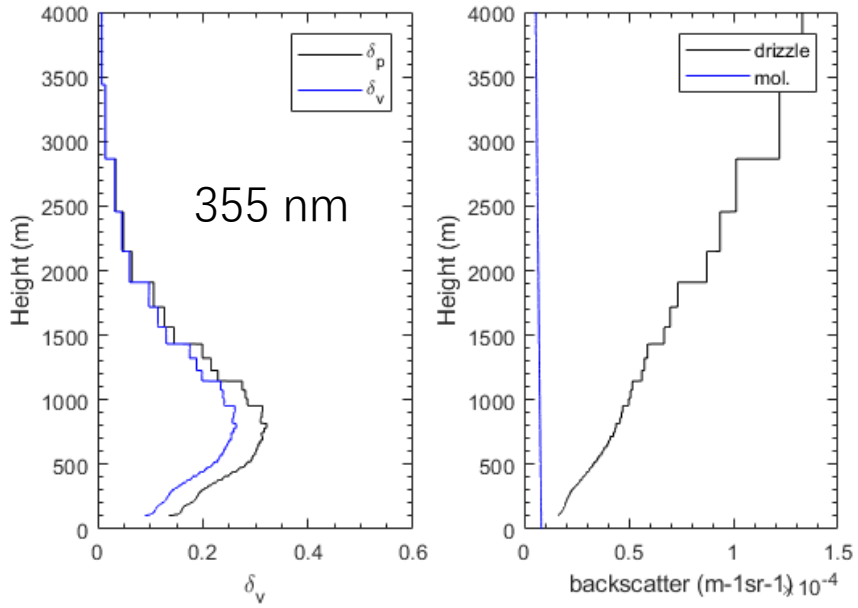
2000:0.1:2001 um



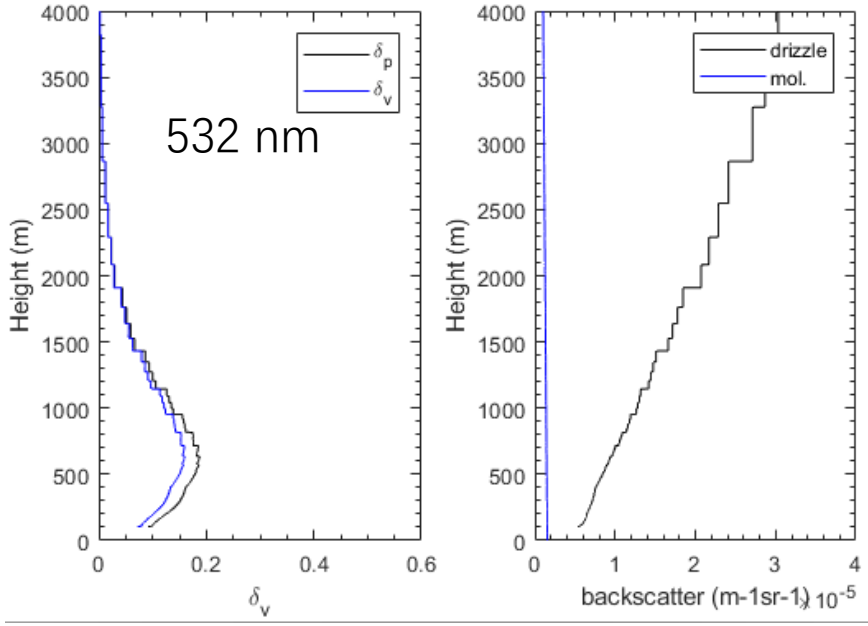


Wavelength Dependent Polarimetry





200 mm telescope
200 mm distance



300 mm telescope
250 mm distance

Scattering by water droplets at VIS band

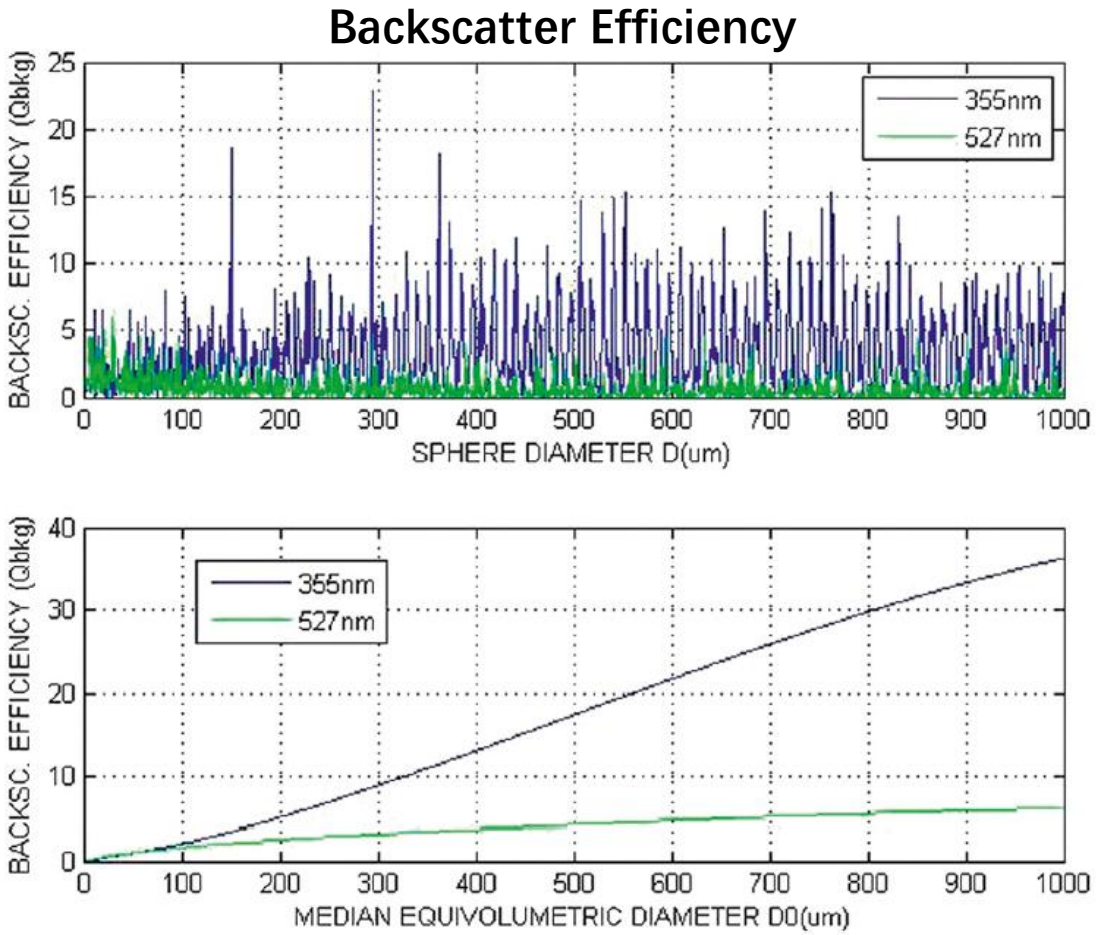


FIG. 1. Lidar Q_{bkg} as defined in Eq. (3), (top) derived from Mie theory for spherical water drops in 1-μm droplet size intervals and (bottom) Q_{bkg} computed over a size distribution defined in Eq. (1) with $\mu = 2$.

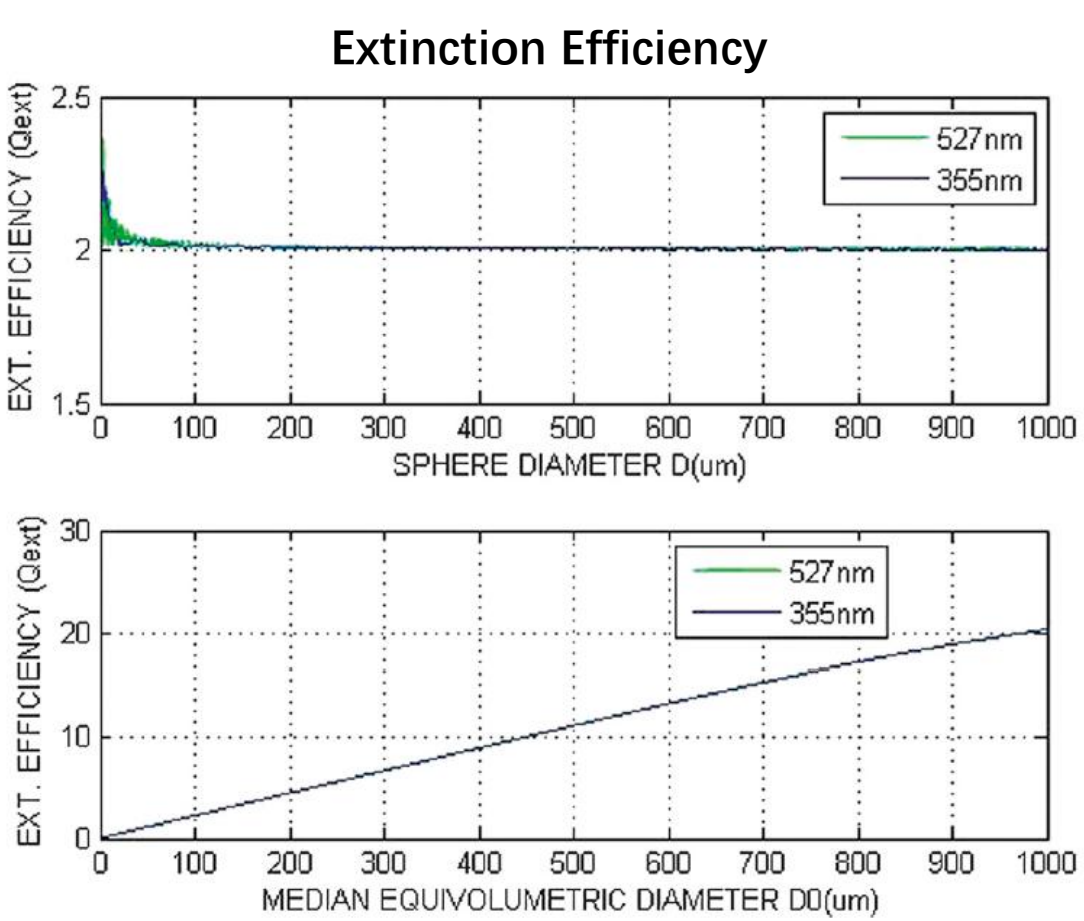
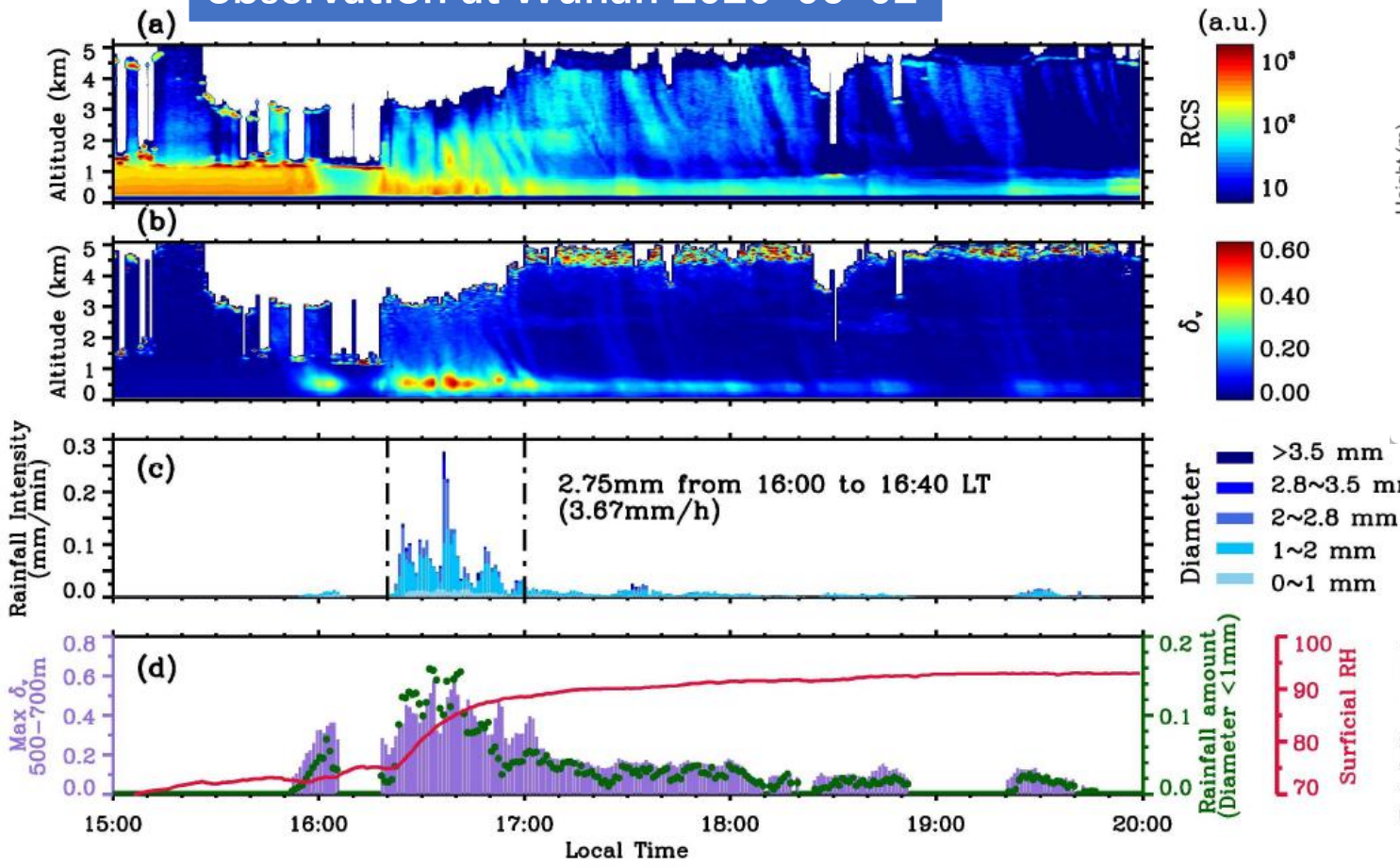


FIG. 2. Corresponding with Fig. 1 Q_{ext} as defined in Eq. (4) (top) derived from Mie theory for spherical water drops in 1-μm droplet size intervals and (bottom) Q_{ext} computed over a size distribution defined in Eq. (1) with $\mu = 2$.

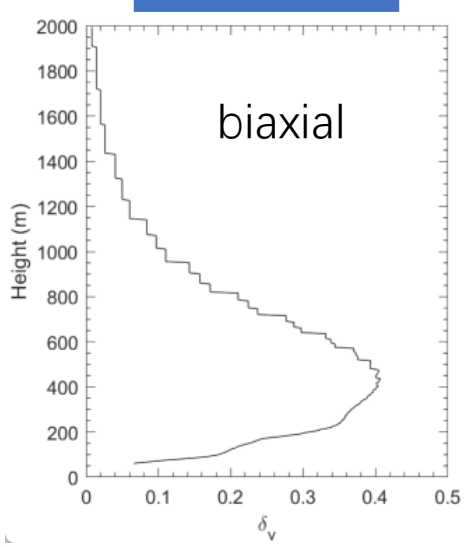
Lolli, S.; Welton, E.J.; Campbell, J.R. Evaluating Light Rain Drop Size Estimates from Multiwavelength Micropulse Lidar Network Profiling. *Journal of Atmospheric and Oceanic Technology* ***2013, 30, 2798-2807, doi:https://doi.org/10.1175/JTECH-D-13-00062.1.***

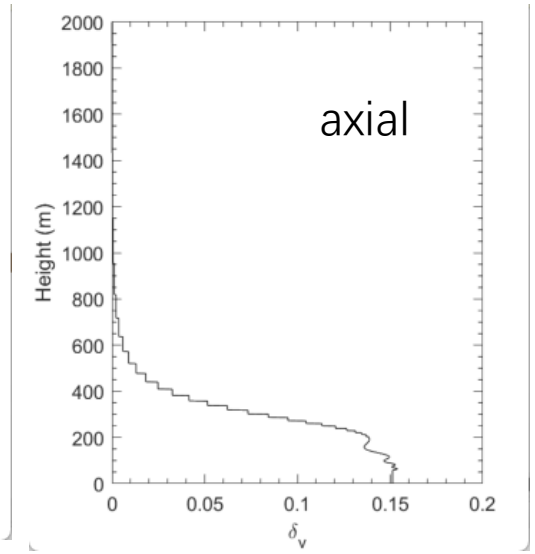
Polarimetry by polydisperse water droplets

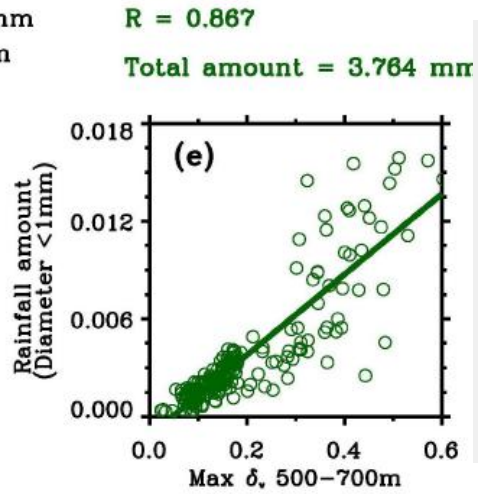
observation at Wuhan 2020-06-02

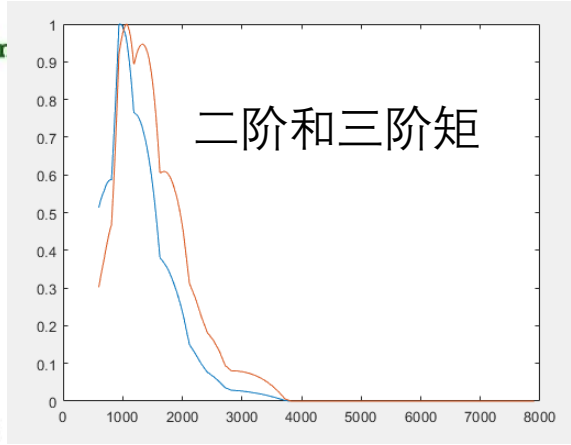


simulation



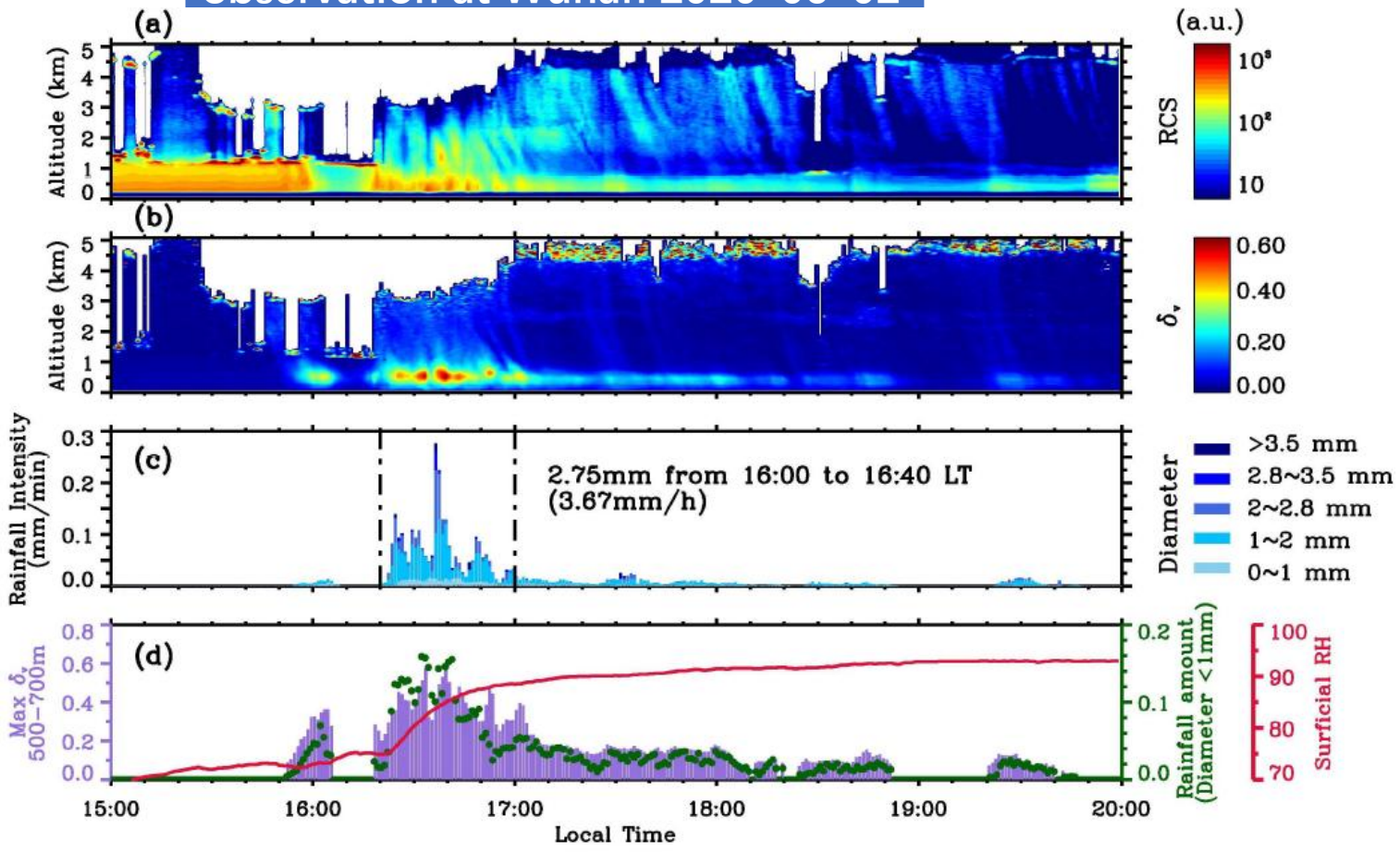




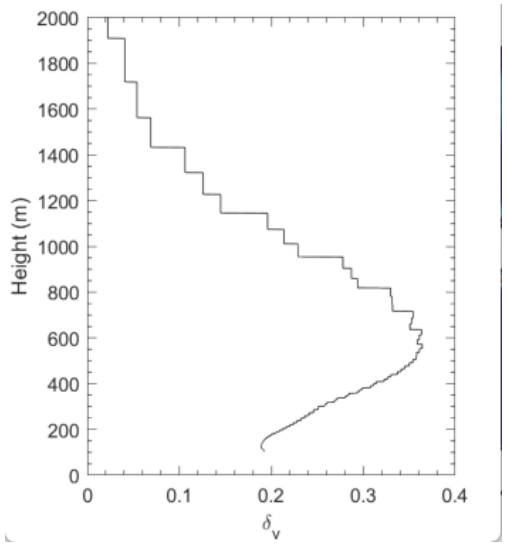


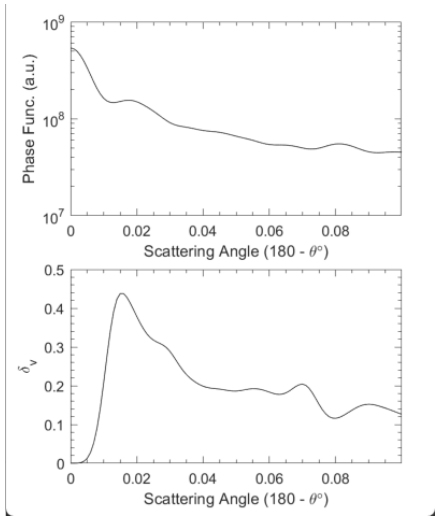
Polarimetry by polydisperse water droplets

observation at Wuhan 2020-06-02



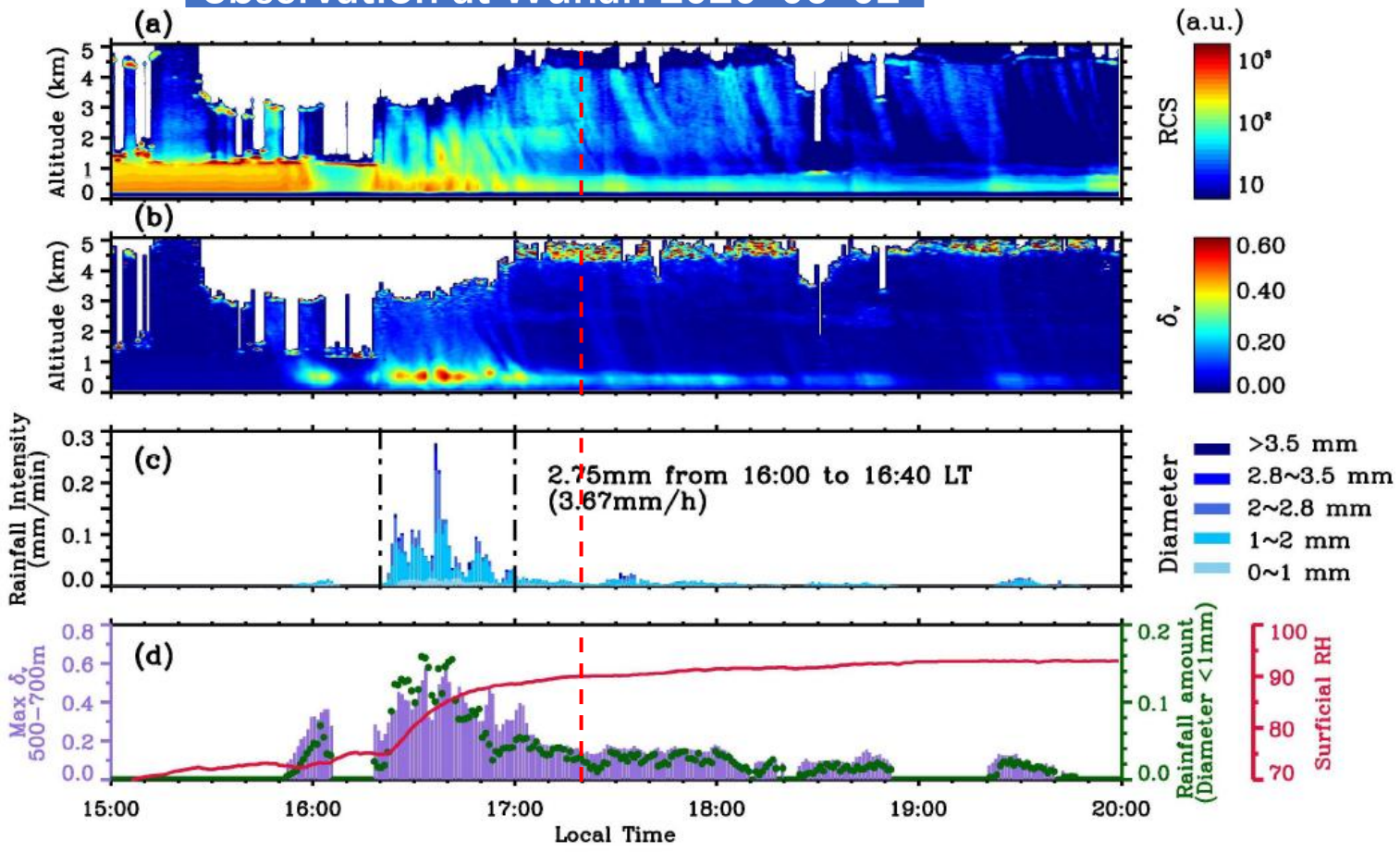
17:00-17:05

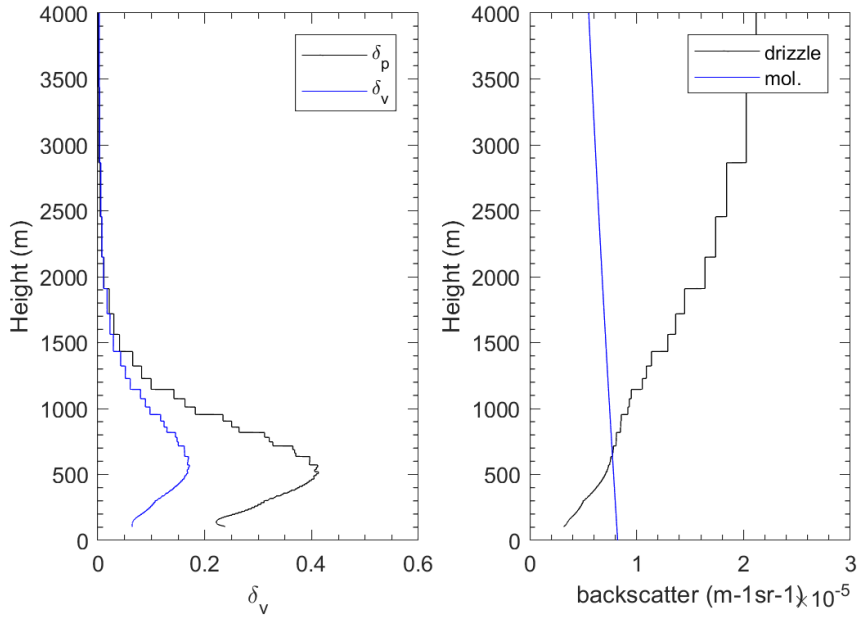




Polarimetry by polydisperse water droplets

observation at Wuhan 2020-06-02





- scattering by drizzles along height
- depolarization along height

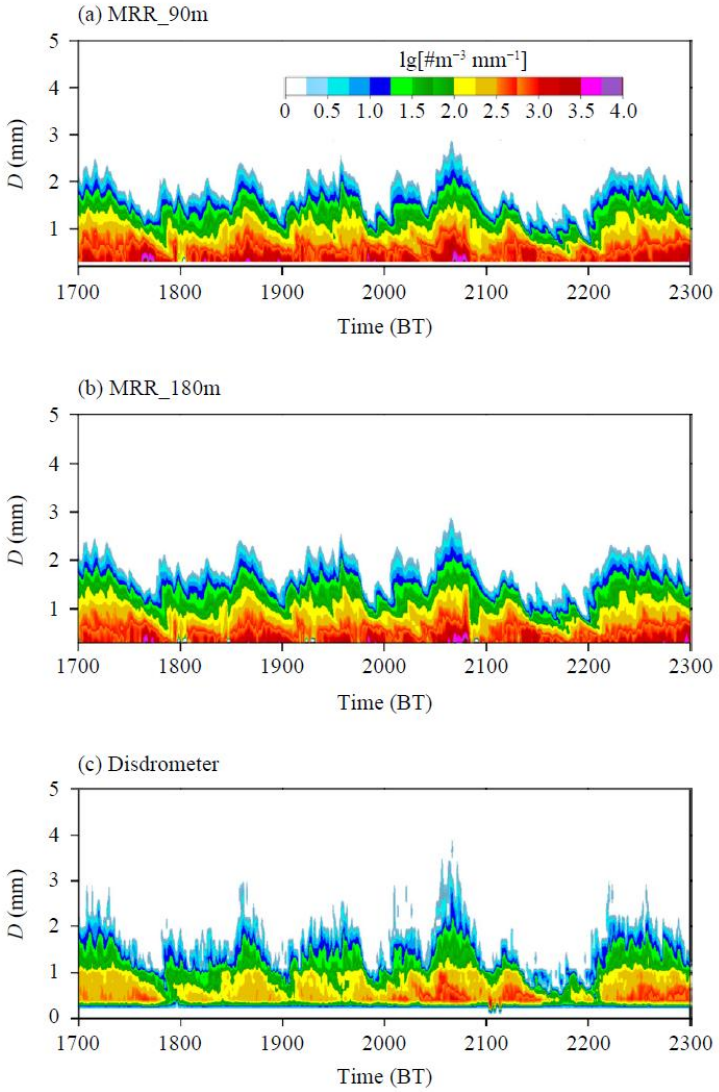
Backup

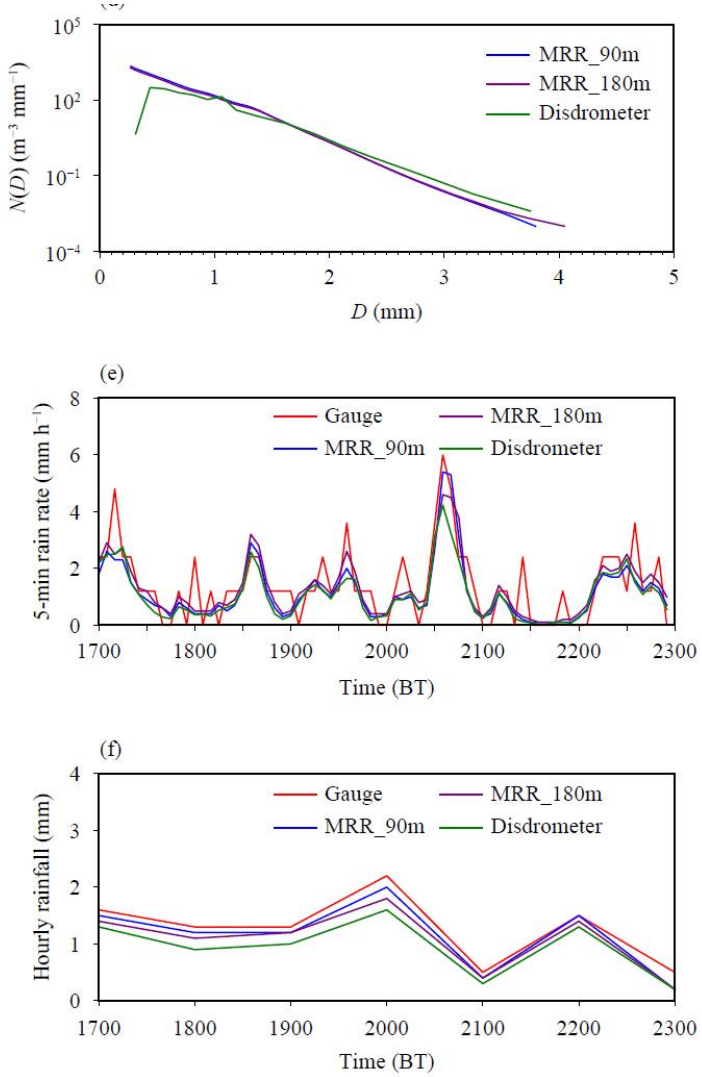
Observed Vertical Structure of Precipitation over the Southeastern Tibetan Plateau in Summer 2021

Gaili WANG*, Renran ZHOU, Jingyi ZHANG, and Ran LI

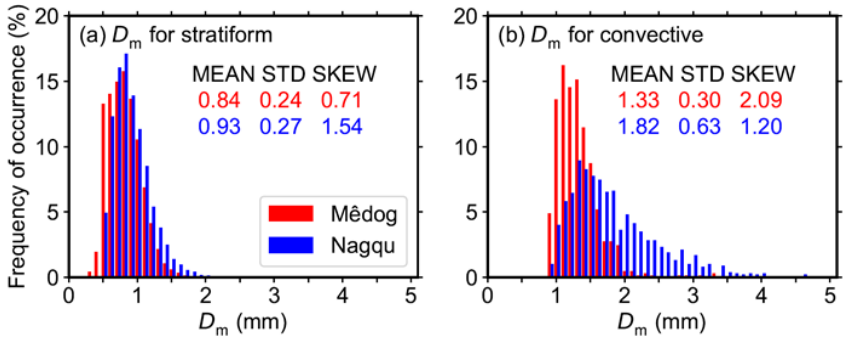
State Key Laboratory of Severe Weather, Chinese Academy of Meteorological Sciences, China Meteorological Administration, Beijing 100081

(Received 5 July 2022; in final form 18 October 2022)

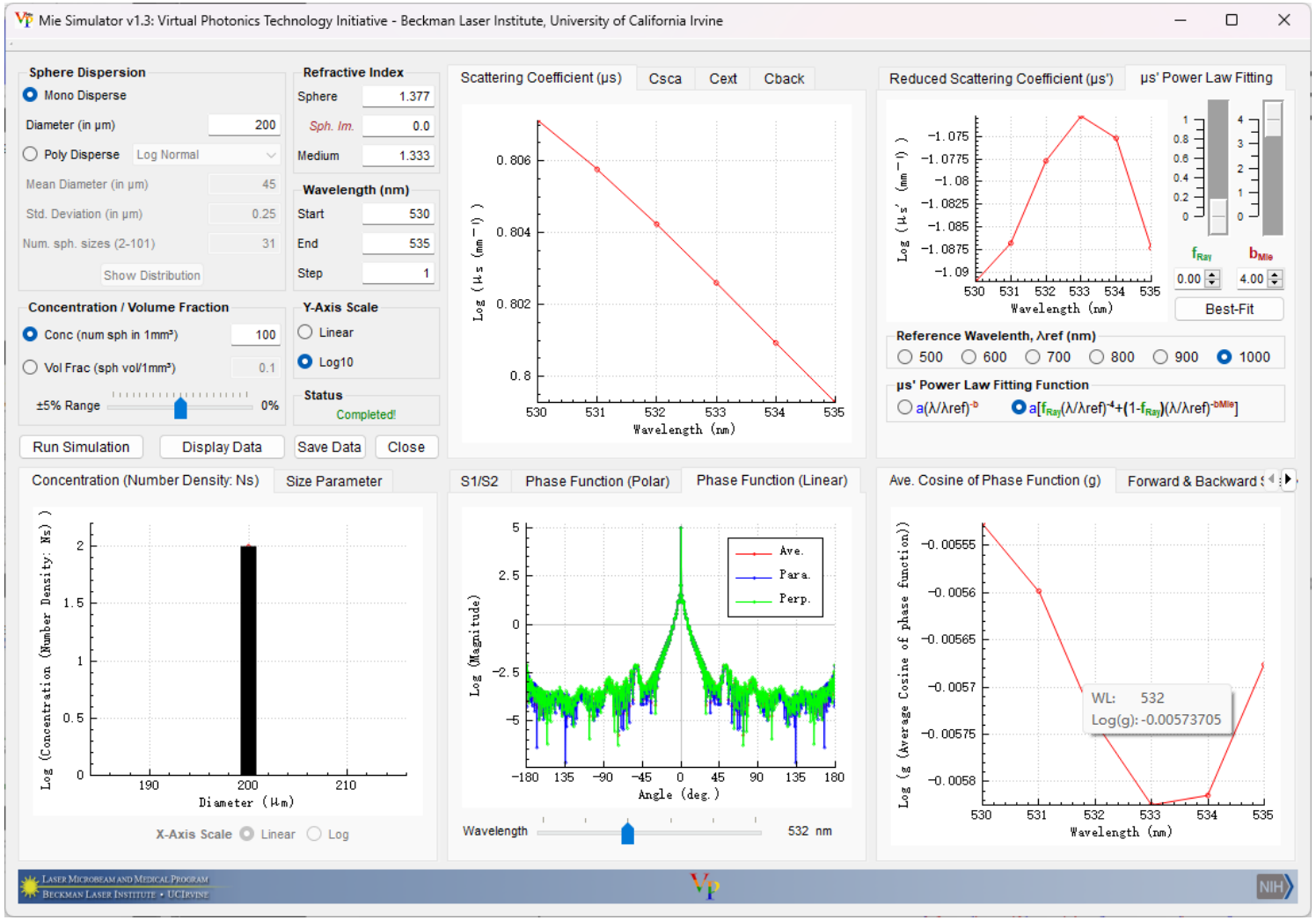


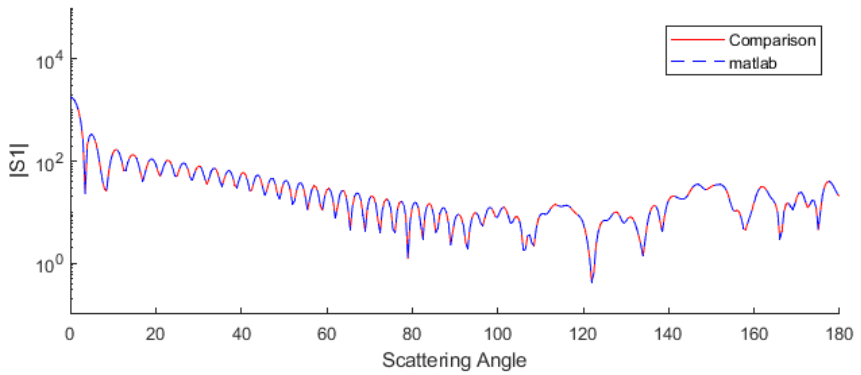


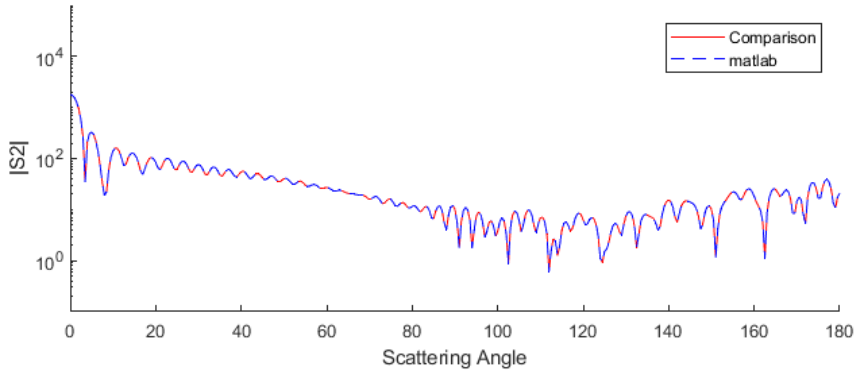
- 整体MRR反演结果跟Disdrometer一致
- MRR小粒径雨滴偏多
- MRR大粒径偏少



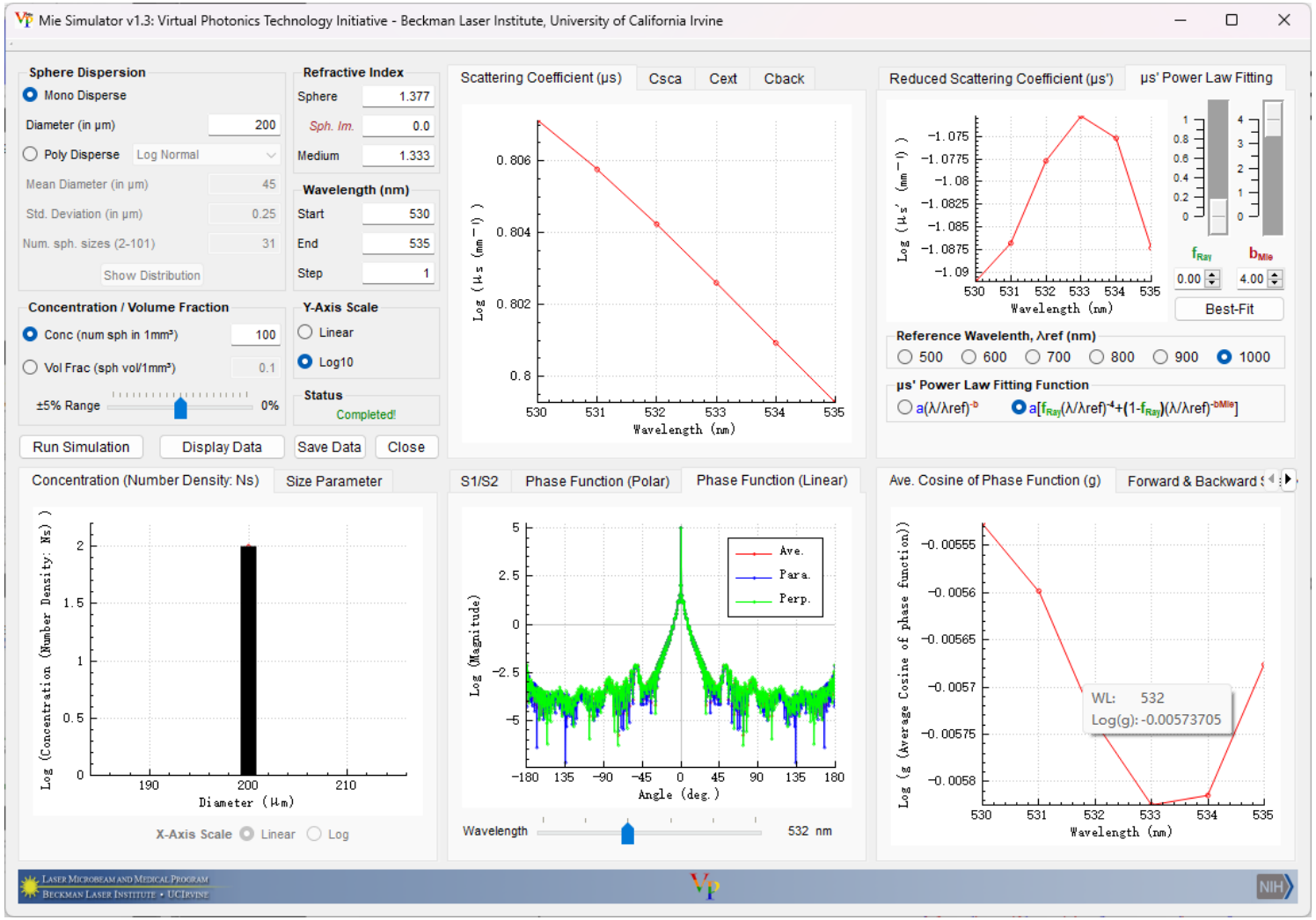
Mie Scattering Code Validation

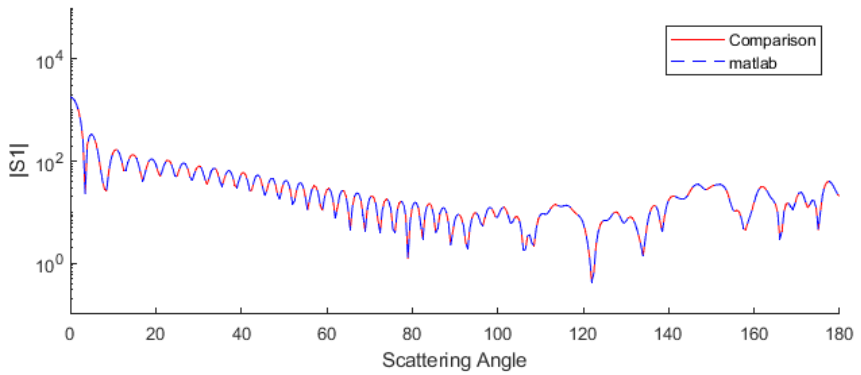


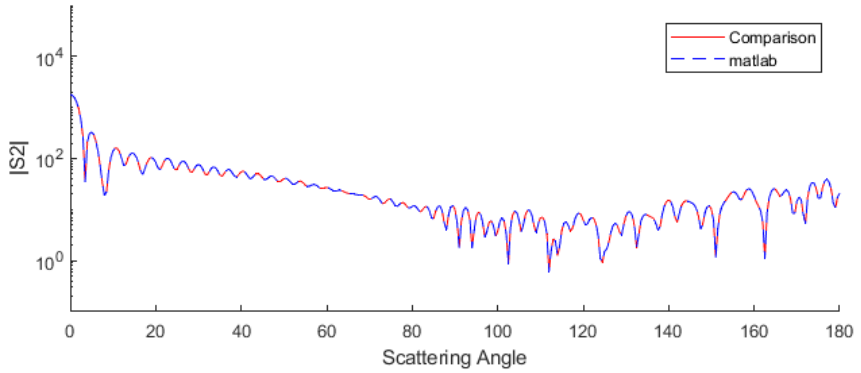




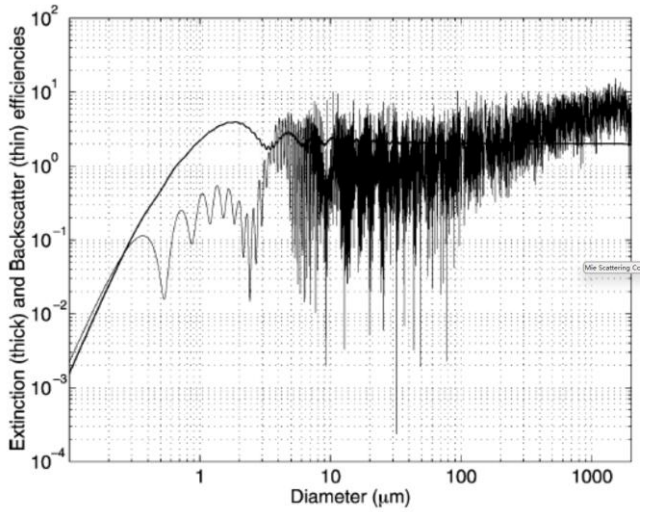
Mie Scattering Code Validation

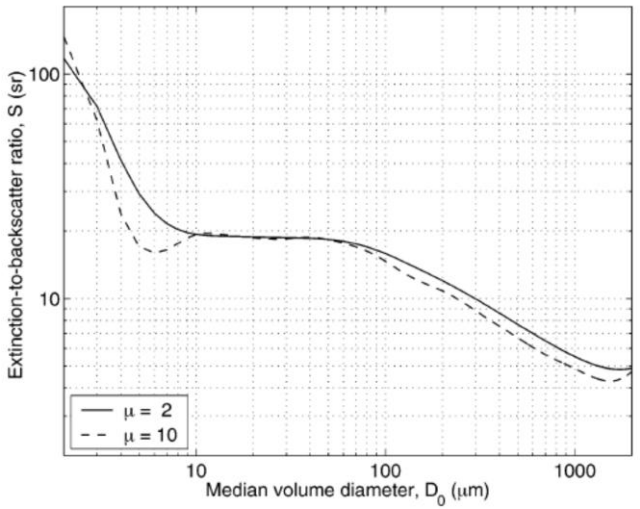






Validation of Mie Scattering Code

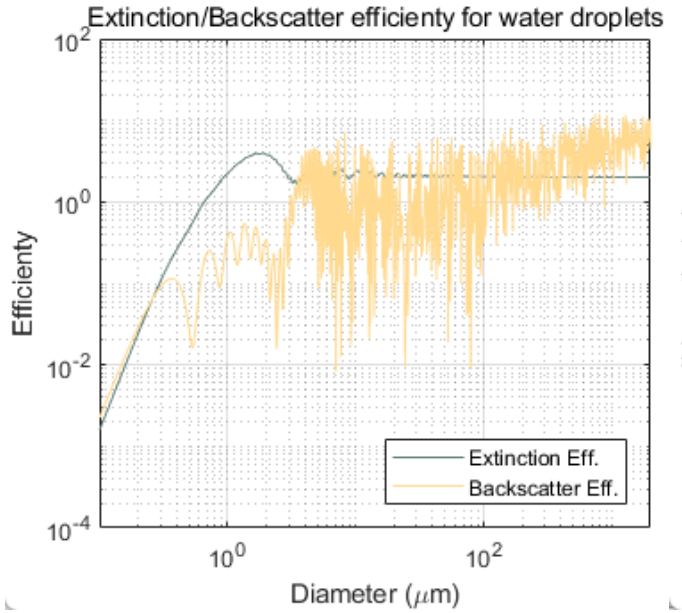


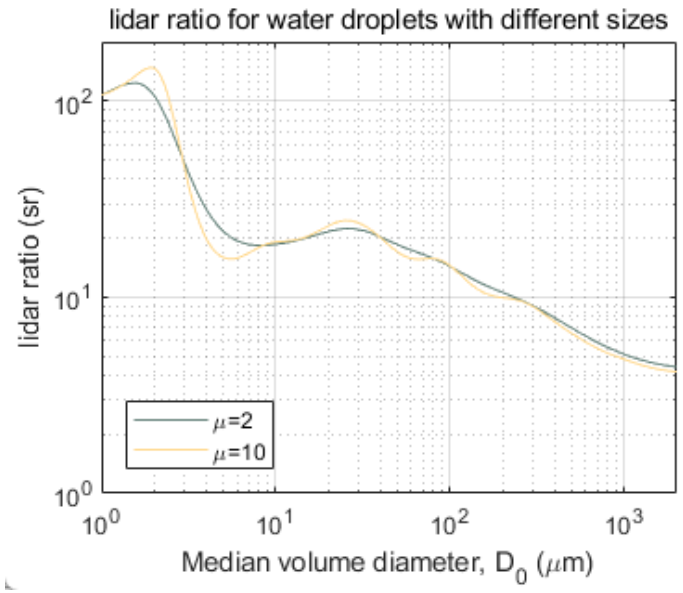


We assume that the cloud droplet size distribution can be described by a normalized gamma function of the form (e.g., Bringi and Chandrasekar 2001)

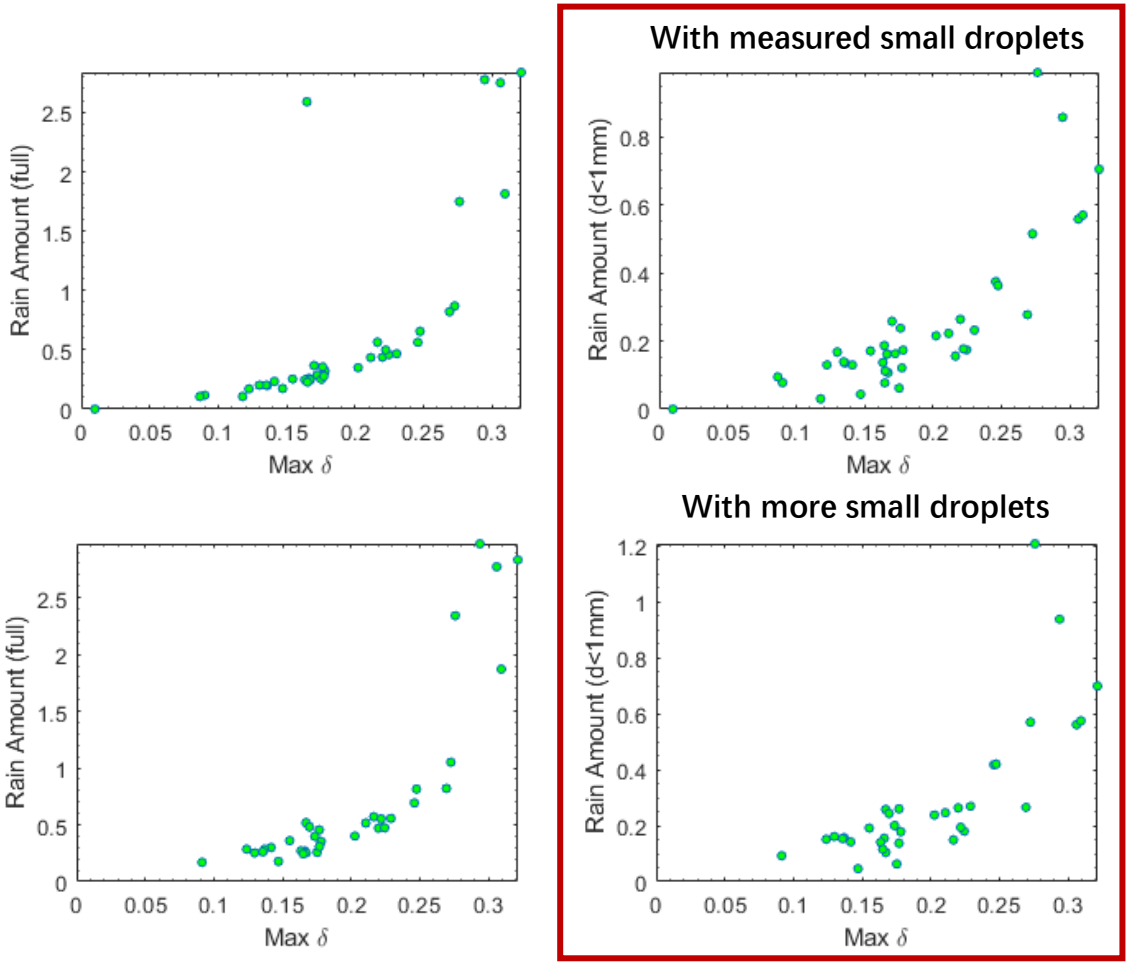
$$n(D) = N_W f(\mu)\left(\frac{D}{D_0}\right)^{\mu} \exp\left[\frac{-(3.67 + \mu)D}{D_0}\right], \quad (7)$$

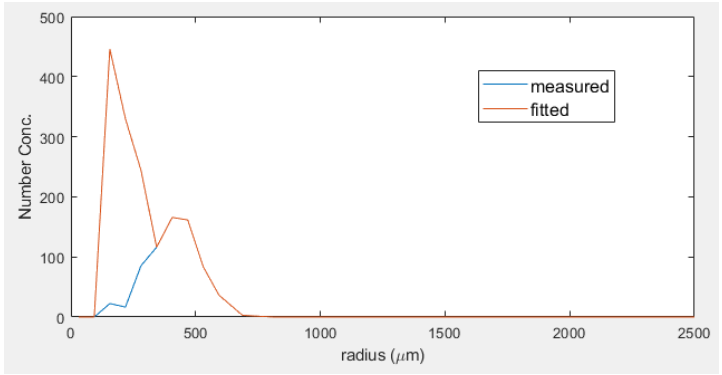
(O'Connor. et al, 2005)

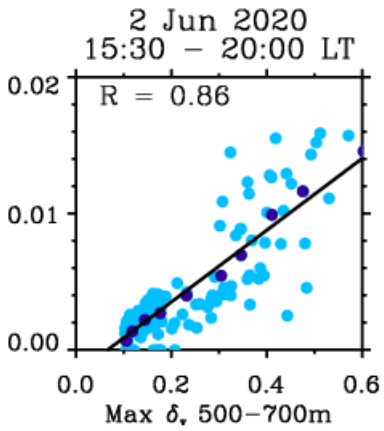




Impacts of small droplets







小雨滴影响较小，小雨滴在低空的退偏信号峰值主要在300米以下，对500-700退偏峰值贡献小